A Comprehensive Survey on Methodologies for Automatic Essay Scoring



R. Johnsi and G. Bharadwaja Kumar

Abstract In the field of educational endeavors, assignments and assessments play an essential position since they allow us to examine student's performance. As of the present moment, evaluation has to be performed by hand, and which requires direct, one-on-one assessment, which is affordable when working with a small class size. But manual grading becomes an overwhelming chore when the number of students grows. Furthermore, prejudice can affect human graders, leading to irregularities and mistakes in the grading process. Numerous scholars have established automated essay grading systems in an effort to overcome these issues. These algorithms take into account various factors such as syntax, semantic coherence, and the contextual use of sentiment-related terms in the text, going beyond conventional keyword based techniques. Hence, this article suggests conducting a thorough analysis of Automated Essay Scoring (AES) models. The primary goal is to elucidate the numerous methodologies, datasets, similarity metric tools, and performance metrics that can be utilized in developing AES systems. Additionally, it discusses the challenges, ethical issues, research outcomes to be considered while developing the AES system.

Keywords Computer-based assessment (CBA) system · Essay scoring · Natural language processing (NLP) · Text processing · Semantics · Linguistics

1 Introduction

Assessment within the realm of education holds substantial importance for evaluating student performance. Traditionally, across most educational institutions, instructors engage in the labor-intensive practice of manually assessing and grading students answer sheets. This conventional evaluation approach is not only time-consuming, but also lacks reliability, making it susceptible to errors. As a result, over the past five decades, researchers have been diligently working on the development of various

R. Johnsi · G. B. Kumar (✉)
School of Computer Science and Engineering, Vellore Institute of Technology, Chennai, India
e-mail: bharadwaja.kumar@vit.ac.in

© The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026

L. Shanmugam et al. (eds.), *Nonlinear Complex Systems, Computational Intelligence and Their Applications*, Springer Proceedings in Mathematics & Statistics 529,
https://doi.org/10.1007/978-981-95-5891-9_47

methodologies to automate the assessment process and replace manual grading with an AES system. AES is defined in various dimension such as

- The process of evaluating and scoring written prose via computer programs [1].
- It can also defined as the task of evaluating one's writing ability and assigning a score to an essay without human interference [2].
- AES is a Computer-Based Assessment (CBA) system that automatically scores or grades the student responses by considering appropriate features [3].

The primary incentive for opting automated assessment system is due to significant disparity in the student–teacher ratio. By automating this evaluation process, assessors can allocate the time they would spend on the assessment tasks to other classroom-based learning activities such as group or class discussions, preparing hands-on activities, class notes, lesson plans, and classroom quizzes or assessments. This benefits for both instructor and their observers. Besides being cost and time management, assessment's subjective nature is an additional significant aspect. When assessed by human graders, this often results in biased scoring and inconsistencies. There may be disparities between the grades provided since various graders apply different grading techniques and have different perspectives. Moreover, some assessors may award higher grades solely based on the positive impression they have on the student. There are instances where an assessor might provide different ratings for the same or comparable responses at various times owing to personal reasons or emotions that vary. In 2023, the statistical report provided by "India Today", the ratio of students to teachers was reported as 24:1. There exists a significant disparity in this ratio among educational institutions in India.

The emergence of the COVID-19 pandemic aligns with a concurrent growth in the use and acknowledgment of AES systems. This pandemic growth has ended up with an integration of essay scoring systems across several schools, universities, and organizations, serving as a significant advancement in the field of education. This integration can be traced back to the reconfiguration of educational delivery approaches by numerous universities, to enhance flexibility, accessibility, and availability for the student community. It emphasizes the increasing significance of AES system. The most key anticipated benefits of AES is to eradicate a human error such as

Rater Fatigue: Fatigue can manifest in various ways, in general it is one of the factors that could negatively affect human performances [4]. Rater fatigue error in essay scoring tasks occurs when individuals are assigned with large number of essays to be examined in a short period of time, they may suffer from mental exhaustion, leading to potentially impact the fairness and validity of essay evaluation process.

Rater Expertise: In essay scoring tasks, rater experience highly influence the overall reliability, accuracy, and validity of essay scoring process [5]. Rater expertise is based on the individual creative skill, subject knowledge possesses when assessing an essay. Training, experience, and ongoing professional development are crucial factors in developing and sustaining rater proficiency in essay scoring tasks.

Severity or Leniency: Severity bias occurs when deserving students fail to receive the credit they deserve, whereas leniency bias is where supervisors prefer to be lenient in their assessments process [6]. Ensuring a balance and establishing standardized scoring criteria is important for maintaining the fairness, reliability, and validity of essay assessments.

Stereotyping: In the context of essay scoring tasks, stereotyping arises when raters permit pre-existing beliefs or biases regarding specific attributes, like gender, race, or socioeconomic status, to impact their assessments of written essays. The authors [7] proved that, on average, girls outperform boys on school exams. In order to mitigate these stereotype issues to promote the blind grading, provide detailed scoring criteria to raters and regular monitor to address any biases in essay scoring process.

Halo Effect: It denotes the inclination of raters to allocate similar ratings across different criteria [8]. This bias poses a challenge to the fairness and precision of essay assessments, underscoring the importance of reducing subjective biases in the evaluation process.

Subjective Nature of Rater: Subjectivity argues that perception emerges from a subject's point of view. Rater subjectivity leads to severity or leniency and reduces assessment validity [9]. The subjective nature of essay assessment leads to variations in scores and error awarded by human assessors, which is perceived by students as a source of unfairness. By providing clear scoring rubrics, training raters, thus enhance consistency and reliability across essay evaluation tasks.

Variation in Response: This challenge arises from the fact that students can provide different responses to a single question posed by an assessor. These diverse answers encompass variations in text length, use of language context, slang, the relevance of the content to the given prompt, the development of ideas, as well as the analysis and evaluation of coherence and cohesion within these responses seems to be a challenging task.

Scalability Issues: As the number of student increases, the scalability of human grading seems to be a challenging task and lacks in providing instant and detailed feedback to students.

AES can effectively address the above shortcomings by ensuring consistent assessments throughout the evaluation process. The majority of currently available AES demonstrate robustness in assessing Multiple-Choice Questions (MCQs) in higher education assessments [10], while the evaluation of short essay type of responses remains a challenging endeavor. The inception of the AES system dates back to 1960, and researchers have consistently contributed innovative ideas to enhance its robustness. Over the last half-century, researchers have delved into a wide spectrum of methods for advancing AES, transitioning from rule-based and statistical methodologies to the cutting-edge advancements of deep learning and sophisticated language models. The rule-based approach offers the advantage of facilitating the integration of domain-specific knowledge with linguistic features, such as Part-Of-Speech (POS)

or syntactic parsing [11–13]. Most of these models have concentrated on the meticulous design of linguistic features, although this process is labor-intensive due to the need for handcrafted feature engineering. Even for human graders, capturing the student's overall intent to assign an effective score remains a formidable task. Deciphering human language is exceptionally challenging from a machine's perspective, primarily because of its complexity. The multitude of ways words can be structured in a sentence, leading to varying interpretations, and the capacity of words to possess multiple meanings in diverse contexts underline the significance of context-based knowledge as a necessary and challenging skill for accurate sentence interpretation.

Despite ongoing efforts, none of the outcomes has demonstrated notable efficiency. This study article focuses on a variety of resources, similarity tools, and approaches, making our study the most thorough and up-to-date assessment on Natural Language Understanding (NLU) evaluation. In this survey paper, we address the 10 analytical investigation questions of AES task. By scrutinizing these questions, the researchers can improve their knowledge in deploying the robust AES system.

2 Glossary of AES System

2.1 Definition of Survey in Terms of Research

A survey paper is a type of research article that provides a comprehensive review and analysis of existing approaches, methodologies, or studies related to a certain topic [14]. Survey papers are important sources of information for researchers because they provide a comprehensive overview of the literature that is currently available and make it easier to understand the trends, developments, and challenges in a particular field of study.

2.2 Why a Survey on AES System

Conducting a survey analysis of essay scoring assignments is a crucial step toward advancing the education field, encouraging cooperation, and improving the general efficiency and quality of AES systems. Due to dynamic nature of AES technology states that older comprehensive review papers may not include the latest advancement systems, thus delivery them less relevant for researchers and practitioners as they may not accurately reflect the current state of the field. Ethical concerns, such as biases in AES systems and their impact on fairness, may not receive thorough consideration in all survey papers. Given the importance of ethical issues in Artificial Intelligence (AI) and education, this can be a significant limitation. In contrast to the aforementioned study, our survey is specifically centered on the assessment of essay scoring system

under NLU, concentrating on the providing knowledge's to new researchers to deploy robust AES systems.

2.3 Recent Works on AES System

2.3.1 English Language

Crowdsourcing-based Automated Essay Scoring (CAES) framework was proposed to tackle the issues while using various deep learning models, which seems to be expensive owing to costs associated with data gathering and model training. This proposed work [15] introduces the Incremental Learning with Dynamic Exemplar Herding (ILDEH) technique, which addresses both catastrophic forgetting and idea drift at the same time. CAES framework gathers data gradually through crowdsourcing and iteratively refines the AES models which requires only 10.6% of the training time required by other techniques. Further investigation showed that a single Central Processing Unit (CPU) server can provide daily updates for over 300 AES models, fulfilling the requirements of the vast majority of real AES systems.

In addition to reduce the limitation of less training data, while using deep learning algorithms the authors proposed data augmentation tool for AES systems using EssayGAN (Essay Generative Adversarial Networks) [16], an automatic essay generator based on GANs architecture to generate a new essays. By fine-tuning the AES system using both real and new generated essays, the performance of the system is improved. Conventional methodologies in AES primarily focus on surface-level linguistic characteristics, disregarding the more profound semantic attributes. In an effort to tackle these limitations, an innovative deep learning approach is adopted [17] which utilizes multi-scale features, employs Sentence Bidirectional Encoder Representations from Transformers (SBERT) to convert sentences into vectors and connect them to the Deep Neural Network (DNN) based AES model. This methodology incorporates shallow linguistic attributes and prompt-related aspects into the distributed features of the essay. The comprehensive approach utilized in the experiment achieves a Quadratic Weighted Kappa (QWK) score of 79.3% on the Kaggle ASAP (Automated Student Assessment Prize) competition dataset.

The authors [18] considered the logical structure for improving the accuracy of AES system. The proposed method estimates the logical structure of each essay using transformer based DNN model with modified self-attention, and output a distributed representation. The distributed representation of the logical structure is then integrated into conventional neural AES models to predict the essay score. The result of the proposed method is demonstrated through experiments using benchmark ASAP data for performance analysis. The comprehensive representation of an essay scoring tasks is enhanced by fusing semantic (dynamic textual semantic representation and topic similarity), linguistic (lexical and syntactic features), and structural attributes of an essay [19]. The model improves the QWK values by an average of 1.4% when tested against state-of-the-art AES models and datasets.

A language model plays a crucial role in deploying AES system. The proposed work [20] utilizes transformer based Generative Pre-trained Transformer (GPT) text-davinci-003 model in AES tasks. The work also explores the influence of linguistic features on AES with GPT, aiming to improve accuracy and reliability of AES by comparing the Educational Testing Service (ETS) corpus of English as a Foreign Language (TOEFL11) to other benchmark level. The authors [21] proposed Bidirectional Encoder Representations from Transformer (BERT) based architecture that captures semantics, rule based, and organizational structure in student response. The combination of these features that conclude feature-enhanced capsule networks improves accuracy and robustness of AES system. The proposed work is validated through experimentations with ASAP dataset and proves to best performance with 81% QWK values than other baseline models. In order to ignore the coherence issues of AES system, the work [22] aims to combine, Bidirectional Long Short Term memory (Bi- LSTM) model over pre-trained RoBERTa (Robustly Optimized BERT) language model, outperformed other existing methods in essay scoring tasks while experimenting with ASAP datasets. Topic-aware BERT automatically retrieves key topical sentences in student essays by probing self-attention maps in intermediate layers of topic-aware BERT which involves fine-tuning a pre-trained language model to learn relations between essay scores, text representations of student essays, and topical information in essay writing instructions [23]. The proposed topic-aware BERT model provides promising results while experimenting with two datasets namely ASAP dataset and manually annotated dataset.

To capture structured coherence and discourse information from student response as graph aggregation. In this proposed work [24], a novel approach using a hierarchical graph structure based on Graph Convolutional Networks (GCN) encodes the hierarchical structure of essays and demonstrating the effectiveness of the proposed work in ASAP dataset and proves to be better than other neural network architectures.

2.3.2 Chinese Language

Many AES works have been deployed by using fine-tuned transformer based models. But selecting the better features for developing robust model seems to be a very challenging tasks. In order to obtain better features, multi-task learning is adopted by designing two auxiliary tasks such as prompt prediction and prompt matching [25]. This proposed work creates a prompt feature fusion method that is better suited for fine-tuning. The experimental results show combination of the two auxiliary tasks with the NEural contextualiZed representation for CHinese lAnguage understanding (NEZHA) pre-trained encoder produces the best results, with QWK improves by 2.5% on average across all results on the Hanyu Shuiping Kaoshi (HSK) dataset.

To show the effectiveness of domain adaptability of essay scorer, multistage pre-trained methods are proposed [26] involving supervised cross-prompt fine-tuning, weakly supervised pre-training, and supervised target-prompt fine-tuning. For weak supervision tasks, an essay scorer model is pre-trained on a large essay dataset. To provide extra supervision, the pre- trained essay scorer model is fine-tuned using

previously rated essays that have the same score range as the target prompt training data. For experimentation of the proposed work, authors collected data from the LeleKetang website written by Chinese school (grade 7–12) students, covering multiple genres and diverse topics of essays such as narrative, argumentative, and prose essays. The essay rating ranges from 1 to 4, indicating poor, normal good, and excellent. The evaluation of Chinese dataset prompts shows that this method can improve a state-of-the-art neural essay scorer in terms of effectiveness and domain adaptation ability and gain moderate performance, with an F1 score of approximately 0.74. Another work to solve the prompt independent model for Chinese second language writing has been proposed [27]. The work combines 90 linguistic features with low-level textual representations to consider both language complexity and correctness, used for the analysis of essays using the Ordinal Logistic Regression (OLR) model. The simple linear model is used to analyze the linguistic features of Chinese L2 AES studies and gained the performance of QWK of 0.714, RMSE of 1.516, and Pearson Correlation Coefficients (PCC) of 0.734.

The proposed auxiliary AES has attracted considerable attention in the field of language assessment, with the objective of enhancing the subjective aspect and improving the precision and effectiveness of the scoring mechanism [28]. This model takes into consideration the lexical inventory and lexicon of the Chinese language proficiency test for individuals living abroad, and is trained and evaluated using a variety of deep learning models. It has been observed that the Convolutional Neural Network (CNN) exhibits the most optimal predictive capacity when applied to its compositions.

2.3.3 Indonesian Language

In order to detect and correct writing errors automatically, the proposed work [28] aims to develop methods that can automatically detect and correct these errors in written responses. The application of Jaro Winkler and N-Gram techniques has demonstrated effectiveness in the identification of writing errors. The final evaluation is conducted using the Vector Space Model (VSM) method, which assesses the similarity between the answer keys and the student answers. The dataset comprises 115 responses provided by 23 participants, each of which may contain writing errors. The outcomes of the Jaro Winkler and N-Gram methods demonstrate a high level of efficiency in detecting and correcting Indonesian words, with an average detection accuracy rate of 83.64%. In future, advanced Natural Language Processing (NLP) techniques are employed to enhance word recommendations.

The proposed work highlights the use of BERT sentence embedding pre-trained models as a richer feature for deploying automated essay-scoring tasks in Indonesian languages. The proposed work [29] has been experimented with using the ukara dataset of the Indonesian language for AES tasks and gained a higher F1-score of 0.829. It surpasses the previous fastText sentence embedding, stacking and XGBoost (Extreme Gradient Boosting) model.

The proposed work improves the quality of Indonesian AES tasks. Modified Long Short Term Memory (LSTM) architecture is called dependency tree LSTM [30]. It helps in identifying the dependency relationship between the words in essays using the Indonesian CoNLL data format. The QWK and accuracy obtained by using modified dependency tree LSTM is 0.68 and 82.5%. Thus, it concludes that the proposed work gains better improvement in QWK and accuracy to the start-of-the-art LSTM model for English AES tasks.

2.3.4 Hindi Language

AES system using NLP-based techniques has been deployed in English work, and exploring AES in low-resource language remains a challenging task. In this proposed work [31], Machine Learning (ML), deep learning algorithms such as LSTM and fine-tuned Multilingual BERT (mBERT) transformer architecture model are compared and gained a similar performance to those of the English domain. Due to Hindi, a low-resource language, there is a lack of corpus collections. The model is trained using a translated ASAP dataset and small-scale real-world Hindi corpus and results show that mBERT achieves the average QWK performance value of 0.852. The proposed also provides insights that integrating linguistic knowledge at a reasonable scale, which may bring further improvement in AES systems in Hindi and other languages.

2.3.5 Arabic Language

The author explores the usage of NLP based techniques such as BERT and Word2vec in Arabic AES tasks [32]. The proposed work has been implemented using two Arabic dataset namely Arabic Dataset for automatic short answer grading (AR-ASAG) and self-collected dataset on official Jordanian History course exam. The usage of BERT is specifically capable in capturing semantics in a text by considering the meaning of words in different contexts. The performance using two Arabic dataset achieved the best PCC of 0.84 and RMSE of 1.003 scores respectively.

2.4 Taxonomy of the AES System

This comprehensive review falls within the domain of quantitative approaches in context of narrative reviews, with a focus on analysis of existing knowledge from diverse sources to offer a concise summary of the current state of knowledge on a specific area of interest [33]. These types of reviews are useful for identifying shortcomings in the current body of literature, showcasing potential for future research, and offering a comprehensive understanding of a specific topic. The utilization of narrative reviews is frequently observed in fields such as medicine, psychology, and

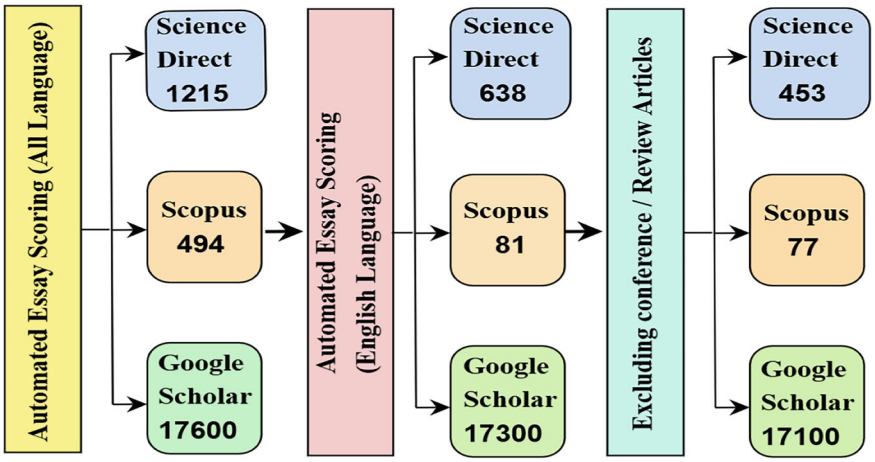


Fig. 1 Taxonomy of the AES system

education, facilitating the consolidation and interpretation of established research findings. This review based on education fields that helps in identifying shortcomings of AES system.

To initiate the Systematic Literature Review (SLR) phase, we conducted an automated search on reputable data science repositories, such as Scopus, Google Scholar, and Science Direct which are visualized in Fig. 1. This comprehensive review refers the publication from the period of 2013 to 2024, shedding light on the diverse advanced techniques employed in AES tasks. The search queries encompass phrases such as "Automated Essay Scoring" or "short answer scoring systems" as well as "Automated Essay Scoring using ASAP dataset," which we conducted across multiple databases.

The above research findings collectively contribute to the development and deployment of effective and reliable AES systems, streamlining the essay evaluation process in educational and testing environments. The above figure also depicts that most of AES work have been done by using ASAP dataset.

2.5 Hierarchical Structure of Analytical Investigations

The hierarchical structure outlined below serves as a general framework for understanding the key steps in bringing a robust AES system in educational fields. The generalized hierarchical structure of Analytical Investigations (AI) used in this survey is illustrated in Fig. 2.

This survey addresses the ten AI questions through which researches can improve their knowledge in integrating AES tasks in NLP educational domain.

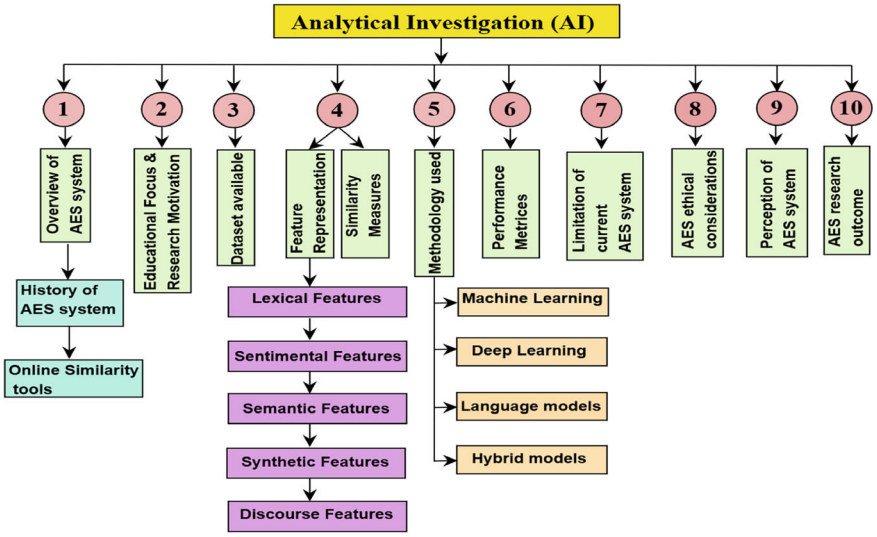


Fig. 2 Hierarchical structure of analytical investigations

3 Analytical Investigation (AIs)

- AI-1: Could you provide an overview of the history of AES systems and sample online semantic similarity tools available?
- AI-2: Do research goals and interests in education motivate studies on AES systems?
- AI-3: Can you specify what datasets are available for an AES system?
- AI-4: What are the feature representation measures available in AES system?
- AI-5: What methods can be employed to make AES more adaptable to different educational contexts and standards?
- AI-6: How do different AES systems and evaluation metrics compare in terms of performance metrics?
- AI-7: What obstacles and limitations exist in the current state of Automated Essay Evaluation (AEE) technology, and what measures can be taken to overcome them?
- AI-8: What ethical considerations, such as privacy and transparency need to be addressed in AES systems?
- AI-9: How do educators and students perceive and experience the use of AES for grading and feedback?
- AI-10: What are the reported research outcomes in automated assessment systems, and what are the future steps for educational applications?

Through these research inquiries, our objective is to gather and consolidate insights regarding the ways in which automated assessment systems can contribute to the improvement and support of both students and educators.

3.1 Analytical Investigation 1

AI-1: Could You Provide an Overview of the History of AES Systems and Sample Online Semantic Similarity Tools Available?

3.1.1 History of AES System

Ellis Batten "Bo" Page Ed. is widely acknowledged as the father of AES, a multi-disciplinary field exploring computer evaluation and scoring of student writing, particularly essays. Page's development of and pioneering work with Project Essay Grade (PEG) software in the mid-1960s set the stage for the practical application of computer essay scoring technology following the microcomputer revolution of the 1990s. PEG employs surrogate indicators to estimate the inherent quality of essays [34]. These surrogates are specific elements related to the writing, like the average word length, essay length, the usage of semicolons or commas, and similar factors. The reliance on proxy measures and formulaic rules in PEG-based essay assessment leads to an inability to capture writing quality in a comprehensive manner.

By 1990, the widespread and substantial increase in power of desktop computers made AES a feasible concept. In partnership with various companies, including Educational Testing Service (ETS) [35]. Page enhanced PEG and conducted effective trials during the early 1990s. Then Intelligent Essay Assessor (IEA) was developed by Peter Foltz and Thomas Landauer. IEA assesses and assigns a score to an essay by employing a semantic text analysis technique known as Latent Semantic Analysis (LSA). IEA may not fully capture the depth of human creativity nuanced expression, and struggle with understanding highly specialized or domain-specific topics. This is because its analysis is based on patterns in a broad range of texts [36].

IntelliMetric represents the AES engine developed by Vantage Learning, with its inception dating back to 1996. Its initial commercial application for essay scoring took place in 1998 [37]. Though this system relies on patterns and semantic analysis, it might struggle with essays on extremely off-topic or those that require deep subject expertise, potentially leading to less precise evaluations in such cases.

In a similar way, ETS presents "e-rater," an AES software has been introduced for a commercial use in February 1999 [38]. ETS utilizes the e-rater engine in its criterion online writing evaluation service to furnish both essay scores and personalized feedback. E-rater primarily focus on surface level features and patterns within the text but fails in capturing content, argumentation, or creativity of the essays. As a result, it might not provide a comprehensive evaluation of an essay's quality, especially in cases where the writing is highly creative or when there are unconventional but valid arguments presented.

Lawrence Rudner has undertaken research on Bayesian scoring and crafted a system known as Bayesian Essay Test Scoring System (BETSY) [39]. Its scoring accuracy may be compromised based on the quality and quantity of the training data.

Its probabilistic approach can be computationally intensive, requiring substantial resources for analysis, which might not be practical in some real scenario.

Over time, AES systems have continued to evolve with an integration of ML, deep learning, and more advanced NLP techniques. These advancements have improved the accuracy and reliability of AES system. Researchers and developers are constantly working to enhance AES systems, improve their accuracy, and make them more adaptable to various writing styles and subjects. Table 1 highlights the different commercial AES system that have been deployed since from 1966. After the 2000s, AES systems experienced significant advancements, leveraging ML algorithms and NLP techniques. These systems analyze various linguistic features, syntax, and content to evaluate and assign scores to essays. Integration of advanced deep learning models further improved accuracy, enabling these systems to handle diverse writing styles and topics. Ethical considerations and ongoing research continue to shape the evaluation of AES system, ensuring fairness and reliability in educational assessments. The widespread adoption of these systems underscores their effectiveness in providing efficient and objective evaluations of written content.

At present, AES have evolved with AI technologies, utilizing advanced deep learning and NLP techniques which incorporates not only adhere to rule based grammars, but efficient in capturing coherence, creativity and contextual understanding of an essay. Improved algorithms consider diverse writing styles, fostering fair evaluations across different genres. Integration of real-time feedback mechanism enhances learning experiences, enabling students to refine their writing skills dynamically. Ethical considerations and transparency in scoring methodologies remain pivotal, addressing concerns about bias and ensuring responsible AI implementation.

3.1.2 Online Semantic Similarity Tools

Semantic similarity tools play a crucial role in AES tasks, aiding in the evaluation of the quality and content of written essays. These tools utilize advanced NLP techniques to assess the level of similarity between student's response and reference text. By measuring the semantic overlap in the language and concepts used, they help educators and institutions impartially grade essays and provide valuable feedback to students. Additionally, semantic similarity tools can identify plagiarism, ensuring the integrity of the assessment process. They enable more efficient and consistent grading, reducing subjectivity in essay scoring, maintain fairness and accuracy in the evaluation of student's writing skills. As technology continues to advanced, these tools are becoming increasingly essential in the educational field, supporting both educators and students in the pursuit of better writing proficiency and assessments.

Despite their numerous benefits, semantic similarity tools for deploying AES tasks comes with several notable drawbacks. One significant limitation pertains to the risk of excessive dependence on these tools, potentially diminishing the role of the human factor in the assessment process. While they excel at efficiently assessing essays for structural and thematic resemblances, they may struggle to capture the full spectrum of coherence and originality present in a student's writing, potentially

Table 1 Comparison between various AES systems

System	Year	Features used	Techniques used	Metrics	Limitations
PEG system by page	1966	Syntax level, grammar, word choice, and sentence structure	Statistical linguistic models (Multiple regression analysis)	QWK	• Relies heavily on surface features • Lack in capturing semantic analysis • Provides no feedback
Criterion online writing evaluation service by ETS	1982	Focus development, organization, style, and conventions	Rule based NLP methods and ML	Precision, Recall, Accuracy	• Limited adaptability to creative or informal writing • Lack of discourse analysis
IEA system by Pearson knowledge text technologies [40]	1998	Semantic analysis features	LSA	Pearson correlation, QWK	• Not effective in assessing the creative writing of an essay • Poor handling of grammar and syntax
Intelli-metric system by Elliot [41]	1998	Grammar, syntax, content relevance, sentence structure	Statistical AI, and NLP techniques	QWK and Pearson correlation	• Not effective for an off-topic responses • Proprietary system with limited transparency
E-rater system by ETS [42]	1999	Grammar, mechanics, usage, organization, style, development	NLP-based feature extraction with statistical scoring models	QWK, Correlation with human scores	• Lacks transparency • Limited semantic understanding • Struggles with non-standardized or creative responses

(continued)

Table 1 (continued)

System	Year	Features used	Techniques used	Metrics	Limitations
C-rater (Conceptual rater) system by Leacock & Chodorow in [43]	2000	Concept-based matching, knowledge representation	NLP techniques, conceptual analysis	Accuracy	• Require more graded essay for training • Focused on short answers • Lacks generalization for broader essay types
BETSY system by Lawrence Rudner [44]	2002	Basic syntactic and stylistic features	Bayesian scoring algorithms	Correlation with human raters	• Limited feature set, struggles with complex writing and deeper semantic understanding
My access by vantage learning's IntelliMetric™ [45]	2002	Grammar, structure, coherence, content	AI-based scoring	Pearson correlation, QWK	• Bias toward training data • Limited adaptability across diverse writing styles
GrammarlyGO	2023	Grammar, clarity, sentence coherence, tone	Transformer-based models (e.g., GPT)	Precision	• Focused on sentence-level corrections • Lacks holistic essay evaluation
WriteLab	2023	Grammar, clarity, sentence coherence, tone	Rule-based NLP with ML algorithms	Precision	• Limited to sentence-level evaluation • Lacks structural analysis
OpenAI codex for writing	2024	Coherence, organization, content relevance	GPT-4 fine-tuned for educational contexts	QWK	• Generalized scoring, lacks domain-specific criteria
Automated insights for essays (AIED)	2024	Coherence, grammar, argument depth	Semantic analysis with advanced machine learning	Pearson correlation, RMSE	• Limited availability, primarily research-focused

(continued)

Table 1 (continued)

System	Year	Features used	Techniques used	Metrics	Limitations
DeepWrite™ by AI innovations	2024	Grammar, coherence, creativity, argumentation	Transformer-based architecture with multi-task learning	QWK, RMSE, Pearson correlation	• High computational requirements • Limited interpretability of scores
NextGen AES (by EduTech Labs)	2024	Structure, reasoning, content relevance	Hybrid approach combining NLP with handcrafted features	QWK,	• Struggles with informal or highly creative writing styles
InsightWriter by adaptive learn	2024	Grammar, sentence variety, content quality	AI-assisted iterative feedback models	Pearson correlation, QWK	• Focused on structured responses • Lacks flexibility for unconventional essays
EssayExpert AI	2024	Grammar, organization, argumentation, citations	GPT-based fine-tuning for educational essays	QWK	• Requires extensive training data for domain-specific tasks
LearnAI WritePro	2024	Grammar, clarity, vocabulary richness	Semantic similarity analysis with contextual embeddings	RMSE, QWK	• Limited effectiveness for handling highly abstract topics

neglecting innovative ideas and unique expressions. Moreover, these tools heavily rely on the quality of the reference texts they compare against, which can introduce biases or constraints into the scoring procedure. Additionally, when confronted with intricate subjects or unconventional language usage, semantic similarity tools may face challenges, resulting in imprecise evaluations. Table 2 highlights sample online semantic similarity tools available for finding the similarity in a text.

Table 2 Sample similarity tools available for finding similarity in a text

Software/Tools	Description	Reference link
SEMantic similarity (SEMILAR) [46]	Helps in finding the word level similarity	https://www.cs.memphis.edu/~vrus/SEMILAR/SEMILAR-screenshots.html
SEMILAT A part of SEMILAR toolkit) [46]	Helps in finding the similarity between short sentences	https://www.cs.memphis.edu/~vrus/SEMILAR/semilat.html
STSAnno [47]	Helps in finding the similarity between text or sentence pairs	https://vukbatanovic.github.io/STSAnno/
University of Maryland Baltimore County (UMBC) [48]	Helps in computing word or phrase similarity in a text	http://swoogle.umbc.edu/SimService/
IndoWordNet [49]	Helps in computing the similarity between two words	https://www.cfilt.iitb.ac.in/indowordnet/
Gene ontology semantic similarity tool (GossTo) [50]	Helps in finding the semantic similarity between genes or terms in the GO	https://paccanarolab.org/gossto/
DIstributionally related words using Co-occurrences (DISCO) [51]	Helps in computing similarity between words or phrases in a text	https://www.linguatools.de/disco/
ProteInOn [52]	It is a Gene Ontology (GO) web tool that helps in finding protein semantic similarity	https://www.lasige.pt/tools/proteinon
Turnitin [53]	I It help educators prevent and detect plagiarism in written assignments, research papers, and other academic work	https://www.turnitin.com/
Grammarly [54]	Grammarly's premium version includes a plagiarism checker that can help you identify instances of text that may be too similar to existing sources	https://app.grammarly.com/
Copyscape [55]	Users can enter a URL or copy and paste text into the tool to search for similarities with content found on other webpages	https://www.copyscape.com/
Plagscan	It helps users identify instances of plagiarism in written documents	https://www.plagscan.com/en/
Quetext	Users receive detailed originality reports that show the percentage of content that may be considered plagiarized, along with the sources of matched content	https://www.quetext.com/

(continued)

Table 2 (continued)

Software/Tools	Description	Reference link
Viper	It helps users detect instances of plagiarism in written documents and content	https://viper.en.softonic.com/
Unicheck	It compares the submitted text to a vast database of academic and non-academic content to check for similarities	https://unicheck.com/
SemSim [56]	It provides similarity scores for words, phrases, or sentences. It uses pre-trained models to measure similarity	https://semsim.xyz/
AI21 labs' semantic similarity calculator	It can calculate the semantic similarity between sentences or phrases. It uses the GPT-3 model for this purpose	https://www.ai21.com/
Luminoso [57]	It offers a simple online tool to measure semantic similarity between words or phrases	https://luminoso.com/

3.2 Analytical Investigation 2

AI-2: What Are the Educational Focus and Research Motivations of Studies with AES Systems?

Studies on automated assessment systems in education focus on enhancing learning outcomes, providing detailed insights into individual student performance, and streamlining the grading process.

Researchers aim to:

- **Improve Learning Outcomes**: AES systems aim to improve learning outcomes by implementing instant feedback mechanisms, facilitating targeted skill enhancement, incorporating adaptive learning paths, integrating formative assessments, increasing student engagement, promoting teacher-student interaction, and aligning assessments with educational objectives. These strategies collectively contribute to a more effective and student-centric learning experience.
- **Individualized Learning Paths**: Automated assessment systems aim to tailor educational experiences by analyzing each student's strengths and weaknesses. This individualized approach helps create adaptive learning paths, catering to diverse learning styles and paces. The integration of individualized learning paths in AES research aims to move beyond a standardized approach to assessment. By tailoring feedback, analysis, and learning resources to the specific needs of each student, these systems contribute to a more personalized and effective educational experience in the context of essay writing.

- **Efficiency in Grading**: Research motivations include developing assessment tools that expedite the grading process for educators. By automating routine tasks, such as checking MCQs or assessing grammar, teachers can allocate more time to personalized instruction and support. Efficiency in grading within the context of AES systems revolves around time saving mechanisms, scalability, consistency, feedback generation, resource optimization, and the promotion of adaptive learning paths. These elements collectively contribute to a more streamlined and effective assessment and learning experience.
- **Validity and Reliability**: Researchers focus on ensuring the validity and reliability of automated assessment systems. This involves designing algorithms that accurately measure not only factual knowledge but also critical thinking, creativity, and problem-solving skills, aligning with broader educational goals.

Ethical Considerations

- Studies explore the ethical implications of automated assessment, addressing concerns related to bias, fairness, and transparency. Researchers aim to develop systems that mitigate biases and provide clear explanations for assessments, fostering trust among educators, students, and other stakeholders.

Thus, the educational focus revolves around improving learning experiences, individualizing education, and increasing efficiency for educators. Research motivations include enhancing the validity and reliability of assessments while addressing ethical considerations to ensure responsible implementation in educational settings.

3.3 Analytical Investigation 3

AI-3: What Are the Datasets Available for AES System?

In recent years, there has been a growing abundance of AES datasets, offering researchers and educators valuable assets to propel both the fields of NLP and educational assessment. These datasets encompass a diverse array of student-written essays, each paired with human-assigned scores that serve as a reliable benchmark for ML models. The accessibility of AES datasets has, in turn, facilitated the evolution and enhancement of automated essay grading systems, simplifying the process of evaluating and delivering feedback on written content on a large scale. Table 3 highlights the different types of dataset available for AES tasks.

Table 3 Different types of dataset available in AES tasks

Sponsored by	Sample dataset	Specificity data peculiarity	Domain covered	Response of students	Sample work	Limitation
Dataset: ASAP dataset **Reference link**: https://www.kaggle.com/c/asap-aes						
William and Flora Hewlett foundations	12,978 samples of essays	On average 150–550 words in length	English, science and arts	Grade 7–Grade 10	[58, 59]	1. Limited prompt diversity 2. Imbalanced distribution 3. Narrow scoring rubrics
Dataset: Mohler dataset **Reference link**: https://web.eecs.umich.edu/~mihalcea/downloads.html						
University of North Texas	2273 answers from 31 students for 80 different questions	Short answer questions	Computer Science	Undergraduate students at the university of North Texas	[60]	1. Limited topic coverage 2. Small sample size 3. Lack of detailed annotations
Dataset: Cambridge learner corpus first certificate in English exam (CLCFCE) **Reference link**: https://ilexir.co.uk/datasets/index.html						
Cambridge University Press and Cambridge Assessment	1,244 exam scripts are present	Full dataset consists of 323,192 sentences	English	Candidate Learners of English	[61, 62]	1. Limited focus on English learners 2. Small dataset size 3. Narrow topic range 4. Lack of detailed feature annotation
Dataset: Argument annotated essays (AAE) **Reference link**: http://b2find.eudat.eu/dataset/ccde0877-a0d0-57e7-9c90-d6a19a63dc5a						
Technical University (TU) Darmstadt	402 persuasive argument student essays	Average sentence length is 4.5	English Argument mining	Created by Stab, Christian	[63]	1. Limited topic scope 2. Small size 3. Narrow domain 4. Lack of detailed annotations

(continued)

Table 3 (continued)

Sponsored by	Sample dataset	Specificity data peculiarity	Domain covered	Response of students	Sample work	Limitation
Dataset: Student Response analysis corpus (SemEval-2013—Task 7) **Reference link**: https://docs.google.com/spreadsheets/u/0/d/1Xe3lCi9jnZQiZW97hBfkg0x4cI3oDfztZPhK3TGO_gw/pub?output=html#						
Beetle II knowledge tutorial dialogue system Science Knowledge tests	BEETLE corpus—3000 student answers SCIENTSBANK corpus –10,000 answers	1- or 2 sentences answers for each question	1. Basic electricity 2. Electronics 3. Life Earth 4. Space science	3rd to 6th grade	[64, 65]	1. Short response format 2. Limited domain coverage small dataset size 3. No deep lingusitic annottaions
Dataset: Response to Text Assessment (RTA) **Reference link**: https://achievethecore.org/collection/18/inspired-by-students-designed-for-change						
Common core state standards	1569 essays and 700 responses	Average number of words is 161.25	Topic: United Nations effort to eradicate poverty in rural village of Kenya	Students with grades 5–8	[66]	1. Limited topic diversity 2. Short essay length 3. Lack of detailed feature annotations
Dataset: Texas dataset **Reference link**: https://data.texas.gov/						
Railroad commission of Texas	87 questions from 2,442 replies	4,450 complaints are addressed	Topic: Introduction of computer science curriculum	Undergraduate students	[67]	1. Narrow focus 2. Small sample size 3. Limited essay length and complexity 4. Lack of detailed scoring annotations
Dataset: UKARA 1.0 challenge dataset **Reference link**: https://nlp.mipa.ugm.ac.id/ukara-1-0-challenge/						
NLP Research Group Universities	**Question one**: 268 responses. **Question two**: 305 responses	Responses to questions contain maximum 2–3 sentences	**Question 1**: About climate refugees **Question 2**: Generic question about people	Indonesian students	[68]	1. Limiated restriction of data 2. Restricted domain 3. Annotation quality 4. Linguistic and cultural bias

(continued)

Table 3 (continued)

Sponsored by	Sample dataset	Specificity data peculiarity	Domain covered	Response of students	Sample work	Limitation
Dataset: SemEval 2013 **Reference link**: https://www.kaggle.com/datasets/azzouza2018/semevaldatadets						
Student response analysis (SRA) in the semantic evaluation (SemEval) workshop in 2013	10, 000 answers to 197 assessment questions	1- or 2 sentences answers for each question	Science domain	3rd –6th grade	[69]	1. Domain-specificity 2. Class imbalance 3. Lack of contextual information 4. Lack of detailed scoring annotations
Dataset: Learning agency lab—automated essay scoring 2.0 **Reference link**: https://www.kaggle.com/competitions/learning-agency-lab-automated-essay-scoring-2/data						
Learning agency lab—Automated essay scoring 2.0 (2024)	10,000 essays	Essays of varying lengths (e.g., 150–550 words)	English, Science and Art	7th–10th grade	Not done	1. Potential scoring bias 2. Limited essay length range 3. Scalability challenges 4. Cultural and linguistic bias

3.4 Analytical Investigation 4

AI-4: What Are the Feature Representation Measures Available in AES System?

This AI aims to discuss different approaches of feature representation. In general feature representation in AES tasks is used to convert the information in raw essay content into a format that can be effectively utilized by various algorithms. Several feature representation methods are employed to determine similarity within a text, such as one-hot encoding, as well as advanced embedding based approaches such as BERT and XLNET. The best way of feature representation helps to analyze the coherence, language proficiency, and overall quality of an essay in an efficient manner. Thus, feature representation plays a vital role in AES tasks.

3.4.1 Basic Similarity Features

Coherence in a text can be analyzed based on content-based features using various similarity measures. It can be analyzed based on word, sentence, or document level using the features extracted from the text. Recent improvement in computational linguistics allows in gathering more linguistic features from an essay. The similarity measures that are considered in AES research are based on lexical level, length-based, word embedding, sentimental, discourse, and ontological features.

- **Lexical Features**

Description: Vocabulary richness is an important indicator for evaluating a student's language level. Lexical based features help us find the vocabulary and ability to use words.

Sample Types of Features Extracted: Uni-gram, Bi-gram, POS tag, stemming and lemmatization, number of characters, words, punctuation, words in quotes, sentences, unique words, hashtags, average word length, and average sentence length, etc., are considered for retrieving the length-based features in an essay.

Examples

POS tag

Sentences: **The old man the boat**

The = DT old = NNman = VB the = DT Boat = NN

Here old man depicts the "old people" and "man" is the verb, meaning that "to operate". Hence, AES system should have to understand this intended meaning and do an evaluation task.

Sentences: Computers don't make people exercise they basically just sit down on their couch and use the computer this won't help to get back into shape. Also,

computers become a habit if a person uses a computer every day of their life it will become an addiction, then they won't be able to stop.

Sample Lexical Features Retrieved

- Number of sentences: 2
- Number of words: 52
- Number of characters: 284
- Average sentence length: 17.3
- Average word length: 5
- Unique words: [Computers, don't, make, people, exercise, they, basically, just, sit, down, on, their, couch, and, use, the, computer, this, won't, help, to, get, back, into, shape, also, computers, become, a, habit, if, a, person, uses, a, computer, every day, of, their, life, it, will, become, an, addiction, then, they, won't, be, able, to, stop]

- **Sentimental Features**

Sentimental features help the teachers to capture the student's opinion or emotional state. For example, in debate type of essay, sentimental words help us to track the author's opinion in text in an efficient manner.

Examples: **Question**: **From ASAP Dataset**

Question: Write a letter to your local newspaper in which you state your opinion on the effects computers have on people. Persuade the readers to agree with you.

Student Response 1: Computers don't make people exercise they basically just sit down on their couch and use the computer this won't help to get back into shape. Also, computers become a habit if a person uses a computer every day of their life it will become an addiction, then they won't be able to stop.

Response: Negative

Student Response 2: Dear local newspaper, I think effects of computers have on people are great learning skills/affects because they give us time to chat with friends/ new people, help us learn about the globe (astronomy) and keep us out of trouble!

Response: Positive
Semantic features

Description: This feature helps teachers to understand the intended meaning in a text.

Sample Types of Features Extracted: Synonyms and homonyms.

Examples

Sentence 1: Computers don't make people **exercise**. They basically just sit down on their couch and use the computer. This won't help to get back into shape. Also, computers become a habit if a person uses a computer everyday of their life it will

become an addiction, then they won't be able to stop. Here, the word "**exercise**" reflects meaning of physical activities.

Sentence 2: The document sets out guidelines on the **exercise** of voting rights.

Here the word "**exercise**" means that something such as a right or choice. Though the word "**exercise**" is present in both the sentences but their intended meaning in a text varies from one sentence to other.

- **Syntactic features**

Description: Syntax analysis helps to find out whether the sentence is grammatically correct or not.

Sample Types of Features Extracted

- **Subject-Verb Agreement**

  Tom rides his bike to work everyday**//Singular subject, Singular verb**.
  These clothes are two small to me**//Plural Subject, Plural Verb**.

- **Superlative Adjectives**

  Jupiter is the **biggest** planet in our solar system**//Superlative adjective: biggest**.

  She is the smartest girl in our class//Superlative adjective: smartest.

- **Coordinating Conjunctions**: Coordinating conjunctions are used to join two main clauses.

  I am driving fast, but I don't think that I will be able to reach on time.

  Main clause one: **I am driving fast**.
  Main clause two: **I don't think that I will be able to reach on time**.
  Coordinating conjunctions: "**but**".

- **Ontological Features**

Description: It helps to analyze the relationship between the concepts or entities in a text. Sample types of features extracted: **hyponyms.**

Expressed in natural language, that "**Delhi is the capital of India**". The sentence has the following parts.

Subject (Resource)	http://www.countries.com/india
Predicate (Property)	Capital
Object (Literal)	Delhi

- **Discourse Features**

Description: Discourse relations (Co-reference relations) specify the relations between sentences or clauses.

Sample Types of Features Extracted: Co-reference features.

Examples

High discourse: "John got home. John gave Mary a call".
Low discourse: "John got home and gave. Marycalled".

Example of Co-Reference Features

John Said Rocky Would Arrive Soon, and He Did.

In the above sentences, the words "he" and "Rocky" refers to the same person. The above said features help to extract meaningful information from an essay to develop a robust essay scoring system.

3.4.2 Similarity Measures

Table 4 highlights similarity measures ranging from simple linguistic features to deep embedding based features to analyze the coherence between reference text and student responses that have been used in AES tasks.

3.5 Analytical Investigation 5

AI-5: What Methods Can Be Employed to Make AES More Adaptable to Different Educational Contexts and Standards?

Several technologies can be employed to make AES tasks more adaptable to different educational contexts and standards starting from ML models to advanced language models. This AIs questions answers recent ML methodologies that are employed in AES tasks.

3.5.1 Machine Learning Based AES System

Numerous ongoing studies in ML for AES or AES tasks predominantly revolve around tasks related to regression or classification. In regression tasks, the objective is to predict the essay's score, while in classification tasks, essays can be grouped into distinct performance categories, such as good, average, or excellent. Since the last five years, most AES systems developed made use of the concept of the neural network and language models. The following research paper highlights the work that have been done in AES tasks using various ML algorithms.

The authors [81] developed an AES system to evaluate approximately 13,000 essays from a ML competition. The essays were divided into eight sets based on different topics, and the focus was on assessing the essay style. They compared

Table 4 Sample NLP based similarity measures used in AES task

Paper	Dataset	Similarity measures	Performance metrics
[70]	ASAP dataset	Pertained GLOVE (Global Vectors) embedding	QWK = 0.768
[71]	Texas dataset	Sentence-Transformers and the OpenAIGPT-3	QWK = 0.70
[72]	ASAP dataset	ROBERTa	QWK = 0.80
[73]	CELA dataset, ASAP dataset	BERT model	QWK values for ASAP = 0.830 CELA = 0.783
[21]	ASAP dataset	BERT	QWK = 0.81
[74]	AR-ASAG, ASAP dataset	Word2vec and Smooth Inverse Frequency (SIF)	The mean squared error for Ara-score dataset is 0.073
[75]	ASAP dataset	BERT	QWK = 0.794
[76]	ASAP dataset	Fast-Text and word2vector	QWK = 0.793
[77]	ASAP dataset	Pre-trained Googleword2vec	QWK = 0.870
[78]	ASAP dataset	XLNET, BERT, BOW model is used XLNET performs well	QWK = 0.75
[79]	ASAP dataset	Glove model	QWK = 0.801
[80]	ASAP dataset	Histogram Intersection String Kernel (HISK) and BOW embeddings	QWK = 0.785
[69]	Essay examination of Computer Graphic course	N-gram and cosine similarity	Correlation value = 0.66

various ML algorithms and found that regression trees achieved the best results with a κ value of 0.52. The author proposed system that [82] designed to address a specific problem or need. It aims to contribute by improving efficiency, enhancing productivity, and increasing accuracy, reducing costs, or solving complex problems. The development process involves research and analysis to understand the problem domain and identify requirements. Designing the system's architecture and user interface follows this stage. Development includes coding using programming languages like Java or Python. Testing ensures that all functionalities work correctly without any bugs before deployment into production environment for user's utilization. Maintenance activities include bug fixing, upgrades, and new feature additions based on user feedback. Overall, the proposed system offers an effective solution through its various stages of research, detailed analysis, designing, coding, testing, & maintenance.

Another work [83] aims to gain the ability to save time, and improve accuracy by combining ML and NLP based techniques to classify and score student answers. This paper deployed ML models such as decision trees, rule induction, Naive Bayes, K-nearest neighbors, and Support Vector Machine (SVM) to classify and score essays, NLP techniques as word2vector to convert word into features and represent in a

semantic space. The models were experimented using ASAP dataset and conclude that SVM with word2vector attains the high precision of (94%) when compared with human score.

The author aims to improve the AES process [83] by using NLP, Word2vec model, and SVM. The authors claim that their method achieves high precision of 94% relative to human resident scores. Precision is a performance metric that measures the accuracy of positive predictions made by a model. In this context, it indicates how well the proposed approach aligns with human scoring in terms of assigning accurate scores to essays. Achieving a precision rate of 94% suggests that the system performs at a very high level compared to human scorers in evaluating and assessing educational goals through essay scoring. To evaluate the effectiveness of Random forest (RF) algorithms, this study [84] provides insights on the effect of RF in non-textual features and text features and compares them with other ML models such as PCC of RF, Gradient-boosted decision trees (GBDT), and XGBoost and concludes that RF model attains the good performance while doing an AES Tasks.

The author introduces a hybrid approach that combines NLP and Random Forest Regression (RFR) [85] for AES. This means that the system uses NLP techniques to extract features from essays, such as analyzing the content and structure of sentences, while also utilizing RFR to predict scores based on these extracted features. Secondly, the paper compares this hybrid model with other well-known techniques in ML like Linear Regression and deep Learning. The performance of each model is evaluated using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE). This comparison helps identify which model produces the best results in terms of accuracy. In summary, the contribution of this proposed system lies in its combination of NLP and RFR for AES purposes. Additionally, it provides a comparative analysis with other models to determine its effectiveness compared to existing approaches. This paper presents an AES system that combines feature extraction models and word vector models to evaluate essays [85]. The goal is to provide a more comprehensive assessment of essay content by considering both numerical features (such as word count and grammar mistakes) and contextual understanding (captured through GloVe word vectors). The system's performance is evaluated using kappa scores, which measure agreement between machine grading and human grading. The results show that the developed system can accurately score most essays, with evaluations close to those of human graders. This approach offers a time- and cost-efficient solution for standardized testing, reducing the need for manual grading while maintaining accurate evaluations.

The Synthetic Minority Oversampling Technique (SMOTE) is a popular approach for improving classifier performance by generating new minority samples and producing a balanced dataset. Enhanced AES may be achieved by sampling the dataset with Gaussian Multi-Class Synthetic Minority Over-sampling Technique (GMC-SMOTE) for imbalance datasets [67]. When conventional classifiers are used on unbalanced datasets, they generally favor the majority class, resulting in poor performance. These systems produce synthetic samples around easily misclassified minority cases, increasing overall classification accuracy while experimenting

with ASAP dataset and particularity evident in essay set 7 where uniform distribution dataset may be necessary. The author proposed [68] work focuses on the integration of prediction scores from various AES models using Item Response Theory (IRT). AES is a process that automatically grades essays without human raters, and different models with different features and architectures have been developed. This proposed work introduces a method that utilizes IRT to integrate prediction scores from multiple AES models, taking into account the differences in their scoring behavior. This method achieves higher accuracy compared to individual AES models and conventional score-integration methods. Additionally, it allows for the interpretation of each model's scoring characteristics and the score-integration mechanism.

Overall, significant challenges persist in deploying machine learning algorithms. While they are effective for straightforward tasks, their accuracy is often only moderate.

3.5.2 Deep Learning Based AES System

A deep learning based AES system employs neural network architectures to automatically assess and score essays. AES systems have shown success in capturing complex patterns within essays, allowing for nuanced and context-aware scoring. However, it's crucial to continuously evaluate and refine these models to enhance their robustness and fairness in diverse educational contexts for several reasons. The following ten recent research papers highlight the work that have been done in AES tasks using deep learning algorithms and conclude that deep learning models outperform ML methods by capturing nuanced and complex patterns in essay structures. Coherence plays an important role in capturing a flow in student essay. This paper proposes a novel approach using sentence embedding BERT along with LSTM and Bi-LSTM to capture the sentence connectivity [86]. This proposed work has been tested on two datasets namely ASAP dataset and Operating System (OS) datasets and achieved average QWK value of 0.76. This paper highlights the importance of coherence and cohesion of essay scoring tasks and also proved the use the sentence level. Embeddings and neural networks help model to capture sequential features and improve the relevance of individual essays. In deep learning based AES, author introduces a multi-level feature fusion model for AES system [68]. The model combines vocabulary-level, sentence-level, and chapter-level features to improve the accuracy of essay scoring. It uses CNN and Bi-LSTM to extract local and global information from different levels of features. The paper discusses the significance of AES in education and highlights the limitations of existing models that focus primarily on vocabulary based features. The experimental results show that the proposed model outperforms state-of-the-art AES models on the Kaggle ASAP competition dataset. The paper also provides an overview of related work in the field and discusses the impact of different levels of features on the scoring model's performance. Overall, the proposed model demonstrates the efficacy of multi-level feature fusion in AES tasks.

To evaluate the impact of deep learning on AES performance, multi-layer perceptron neural networks are employed [87]. By adding more hidden layers to these networks during experimentation with different architectures for explanation models (such as SHAP implementations), it is found that descriptive accuracy increases by approximately 10%. Overall, through their methodology involving deep learning techniques applied to Grade-7 essays' writing style prediction along with evaluation experiments utilizing neural network architectures for explanation modeling purposes, this proposed work contributes toward enhancing our understanding and trustworthiness in AES while also improving its interpretability and potential pedagogical value. To evaluate the effectiveness of RNN (Recurrent Neural Network) model on AES task to learn the relationship between the essay and it's assigned scores without the use of feature engineering [88]. The inclusion of small number of features along GloVe, an unsupervised learning algorithm, is employed to enhance the performance of the system. The experiment was conducted using ASAP dataset and achieved the higher performance than other baseline models. It addresses the drawbacks of grading essay exams, such as cost and fairness concerns by predicting both analytic scores and overall scores, aiming to improve the interpretability of score predictions while maintaining accuracy. The proposed work [89] utilizes a neural approach for AES. It incorporates a Multidimensional generalized Partial Credit Model (M-GPCM) to estimate the latent abilities of the essays. The model includes various layers such as embedding, convolution, attention pooling, recurrent, and concatenate layers. It also incorporates handcrafted features and uses a dense layer to obtain the latent ability vector. The M-GPCM layer then calculates the score probabilities for each analytic scoring item. Finally, the obtained probabilities are used to predict the analytic scores of the essays.

In the proposed architecture [90] for AES tasks, RNN and CNN play important roles in an efficient manner. The multi-channel convolutional layer in the architecture learns and captures the contextual features of the word n-gram from the word embedding vectors. This layer uses different window sizes and performs convolutional operations to extract relevant information from the input. On the other hand, a variant of RNN called Bi-Gated Recurrent Unit (BGRU) is used to access both previous and subsequent contextual representations. The BGRU model is responsible for processing the sequential information in the essays and capturing the dependencies between words. By combining the capabilities of CNN and RNN, the proposed architecture can capture both local and global contextual information in essays, leading to a more comprehensive semantic representation. This enhanced representation can then be used for scoring essays accurately. Overall, the recurrent networks and convolutional neural networks in the architecture contribute to capturing contextual information and improving the automatic essay scoring process. The proposed paper [91] contributes by implementing and comparing three AES algorithms using deep learning models. It also introduces a new algorithm for generating feedback based on Constrained Generation by. Metropolis–Hastings sampling (CGMH), which produces semantically-related sentences. Additionally, the paper builds a classifier that integrates both essay scoring and feedback generation into a systematic framework. The findings support the feasibility of an integrated system that combines AES

with meaningful feedback generation, potentially leading to further advancements in linguistic feature analysis and improved accuracy in scoring essays as well as creating more relevant feedback through larger corpora.

The paper highlights the need for explainable AI models in AES [92] and focuses on developing a deep learning approach using multi-layer perceptron neural networks to improve the performance of AES. The study aims to understand the decision-making process behind scoring essays and evaluates how deep learning affects trustworthiness in explanation models. Additionally, linguistic indices are used to assess writing quality in Grade-7 essays. It is done by training models to predict both holistic scores and rubric scores using datasets from ASAP essay datasets. Feature selection is applied on a pool of 1592 linguistic indices that measure aspects such as text cohesion, lexical diversity, lexical sophistication, syntactic sophistication, and complexity. The impact of adding more hidden layers to neural networks is evaluated by assessing descriptive accuracy improvements. Overall, this work combines deep learning techniques with linguistic analysis methods to enhance AES systems while providing insights into their decision-making processes through explainable AI approaches.

In order to improve the effectiveness of domain adaptability, this proposed work [86] designed the domain-adaptive framework which captures the shared knowledge across different domains. This is achieved by training the model with small well-designed perturbations obtained through a variation of the Fast Gradient Sign Method (FGSM) on ASAP dataset and attains the enhanced model's robustness for domain variation problems. This proposed versatile solution can be successfully applied to different models for improving AES performance across various domains. The author [20] presents a new approach for AES called Siamese Bidirectional Long Short-Term Memory Architecture (SBLSTMA). AES aims to automatically evaluate and score student essays, reducing manual workload and providing faster feedback. The paper addresses the limitation of existing AES methods that only consider the essay itself without considering the rating criteria behind it. To overcome this, the authors introduce rating criteria representation using sample essays provided by domain experts. They create a new input pair consisting of an essay and a sample essay representing the rating criteria. The SBLSTMA model is designed to handle this new input pair. It utilizes Bi-LSTM units to capture both semantic features in the essay as well as information about specific rating criteria represented by samples. Furthermore, a self-feature mechanism is introduced at the LSTM output layer to compute similarities between sentences within an essay and between an entire essay and its corresponding sample. This enhances performance on longer and more complex essays. Experimental results using ASAP dataset demonstrate that SBLSTMA outperforms previous neural network AES methods, showing promise for enhancing automated scoring in education.

3.5.3 Language Models Based AES System

Language models can serve as powerful feature extractors, capturing intricate linguistic patterns, semantic relationships, and contextual information from essays. The embeddings generated by language models to represent the content, style, and structure of essays for subsequent scoring are utilized. Language models can contribute to the development of adaptive scoring criteria, allowing the model to dynamically adjust its evaluation based on the content it encounters. The following seven research papers highlight the work that has been done in AES tasks using advanced deep learning algorithms as language models. Language models deliver the highest accuracy, excelling in tasks involving complex, lengthy essays.

In the field of education notably in AES tasks, pre-trained models like BERT have not been effectively utilized to outperform other deep learning models such as LSTM. This author introduces a new approach that jointly learns a multi-scale essay representation for BERT [75], along with employing multiple losses and transfer learning from out-of-domain essays, resulting in improved performance. Experimental results demonstrate that this approach achieves state-of-the-art results among deep learning models in the AES task and generalizes well to long-text tasks. The author pays the way and provides attention to two language models such as BERT and XLNet [78] to traditional methods like bag of words and LSTM networks for automatically scoring essays. The authors explain the network architectures of BERT and XLNet using clear notation and diagrams, and show that the transformer architectures used in these models outperform traditional recurrent neural network architectures. This study [93] designed a online discuss forum to generate an assessment between essays using pre-trained Sentence Transformers that captures the high coherence between the student response and rubrics. The comparisons have been made using different monolingual and multilingual models and found that distiluse-base-multilingual-cased-v1 model acquires the higher performance of correlation value of 0.63 and MAE score of 0.70.

This paper [20] discusses the use of GPT in AES and explores the inclusion of linguistic features to enhance the accuracy of AES. The study analyzes the scores of 12,100 essays based on three levels (Low, Medium, and High) and compares different regression models to assess the performance of GPT and linguistic measures. The findings suggest that AES using GPT can be carried out with some degree of accuracy and that the combination of GPT and linguistic features improves the prediction of benchmark levels. The paper [92] presents an automated scoring approach for reading comprehension using in-context BERT tuning. The approach leverages item linkage and meta-training to create a shared scoring model for all items, addressing limitations of existing approaches. Experimental evaluation on the NAEP challenge dataset demonstrates the effectiveness of the approach, outperforming other baselines. The authors conduct qualitative error analysis and explore fairness aspects. They provide avenues for future work to improve applicability and fairness. Overall, the paper contributes a novel approach to automated scoring and highlights the potential for further enhancements.

Another work is proposed by author [94] using language models such as R2BERT to enhance the performance of AES by fine-tuning pre-trained language models using a combination of regression and ranking. The authors address the limitations of previous approaches by utilizing the BERT model to learn text representations and applying mean square error loss and batch-wise ListNet loss to constrain the scores. The experimental results on the ASAP dataset show that R2BERT outperforms state-of-the-art neural models and achieves better performance on narrative prompts. The authors also visualize the attention weights of the self-attention mechanism in BERT to demonstrate its effectiveness in capturing conjunction words and key concepts in essays. Overall, the proposed method shows promising results in improving AES performance. The author proposed this work [94] to improve the accuracy of essay scoring by incorporating multi-scale semantic features. It addresses the limitations of traditional AES techniques that only focus on shallow linguistic features and ignore deep semantic features. The method utilizes a combination of document-scale and sentence-level features extracted using LSTM-MoT and SBERT respectively. It also includes shallow linguistic features and prompt relevance features. By fusing these diverse semantic features, the proposed method achieves better accuracy in scoring essays compared to the baseline model. It takes into account the deep semantic meaning of the essay, captures coherence and organization, and considers the relevance between the essay and the corresponding prompt.

The author proposed system [89] that uses techniques from corpus linguistics, computational linguistics, and ML to grade essays. It aims to provide a more efficient and objective way of grading essays compared to manual methods. The system focuses on evaluating specific language features called Coh-Metrix for developing prompt-specific AES scoring models. These models are created by reducing the dimensionality of nine informative features through advanced techniques. To validate the performance of these models, a three-staged scoring framework is used where machine scores are compared against ratings provided by human raters as a benchmark. Overall, this automated essay system improves efficiency and objectivity in grading while ensuring its validity through comparison with human ratings.

Data augmentation in general has the capacity of increasing the size of data by generating the real dataset without actually collecting the data. It plays an important role in evaluating an essay of multiple topics, even with a limited amount of data in essay scoring tasks. Transformer model using transfer learning concepts helps to capture the coherence problems in AES tasks. This proposed work [91] addresses the challenges on evaluating the essays on multiple topics with limited data by combining the data augmentation techniques combined with transformer model such as BERT and RoBERTa. This study improved generalization ability of the models on experimenting with ASAP dataset to evaluate essays across different topics, making it more robust and effective on experimenting with ASAP dataset. Transformer models such as BERT, Roberta fails in AES system because of the limited sequence length. But the proposed model addresses these issues of evaluating the quality of longer documents by developing three models such as Longformer, RoBERTa, and Decoding-enhanced BERT (DeBERTa) and two ensemble models based on Longformer, RoBERTa, and DeBERTa. These models were trained on ELLIPSE corpus and result concludes that

ensemble models, particularly the DeBERTa-Longformer ensemble, outperformed other models and capable of capturing the coherence in an efficient manner [95]. This study conducted a systematic literature review to evaluate the effectiveness of AES systems in real-world education over the past six years [96]. The review analyzed AES algorithms, feature extraction methods, evaluation metrics, and limitations, identifying areas for improvement. This study explores the use of the GPT-3 text-davinci-003 model for AES of 12,100 essays from the ETS Corpus of Non-Native Written English (TOEFL11). It compares GPT-generated scores with benchmark levels and examines how linguistic features impact scoring accuracy. The results show that GPT-based AES is reliable and accurate, offering valuable support for human evaluators [97].

3.5.4 Hybrid Models Based AES System

To enhance the accuracy and reliability of AES system, some hybrid models have been proposed by combining different approaches and techniques. The below research paper highlights previously employed methodologies through the utilization of hybrid models. The analysis concludes that Hybrid systems provide the highest reliability and accuracy by integrating the strengths of multiple approaches.

The author employed hybrid model in AES tasks, by combining transformer based models such as BERT along with handcrafted syntactical features [98]. In order to provide generalizability, the model is trained on multiple essay topics that help in essay responses from various topics. The usage of BERT model helps in maintaining the interaction of words and considering the dependencies between words in essays. The usage of handcraft features helps in analyzing the structure of essays. The combination of BERT along with handcrafted features is validated by using QWK as performance metrics thus concludes that it helps in enhancing the quality, reliability of AES tasks. The proposed work [67] uses the attention based hybrid model for AES tasks that outperforms the other multiple baseline models. It uses deep learning models such as CNN and Bi-LSTM to capture the semantic features in question and answers in an essay and employs the attention mechanism to determine the relative weight of essays to obtain the coherence. Then finally the model combines all feature vectors and input them to network model to obtain the corresponding essay score proved to gain a good performance. To eliminate the need of feature engineering, deep learning models are used and conventional handcrafted features are used to analyze the structure of essays. This proposed hybrid model integrates handcrafted features with many deep learning models such as LSTM, Bi-LSTM, and BERT [99]. In this case, the model incorporates 25 various handcrafted essay features which are based on length, syntax, word, and readability based features. By experimenting with ASAP dataset, BERT with handcrafted features attains the higher performance than other baseline neural network models.

The author designed a framework, for prompt independent AES where rated essay for target prompt are not available. The proposed [100], Two-stage Deep Neural Network (TDNN) address these challenges using shallow model in first stage and

end-to-end hybrid deep learning model in second stage. The sentence structure is analyzed by using syntactic network with semantic or POS features. The proposed model was implemented by using ASAP dataset and attained the higher performance than other prompt dependent AES models. To reduce the bias in grading, the proposed model aims to design the Intelligent NLP Essay Grading System (INLPEGS) [101] that are designed based on enhanced LSA using POS n-gram Inverse Document Frequency (IDF). The model have been experimented using two datasets namely, ASAP and a dataset of Computer Science questions and answers from Babcock University and result shows an accuracy of 89.03% when compared human score with an automated model score.

The author proposed a grading model that addresses the issue of time-consuming and can lead to subjective judgments and unfairness in the evaluation process [102]. The proposed model proposes an automatic essay grading model that combines multi-channel convolution and LSTM. The model incorporates a dense layer, softmax function, multi-channel convolutional neural network, and LSTM to extract and model text features of different granularities. The model is experimented using ASAP dataset and concludes that 6% improvement in automatic scoring effectiveness compared to a strong baseline model. The author proposes a hybrid approach that combines NLP and neural network models [103]. The proposed system used writing features analysis that relies on a set of pre-graded essays to assign grades. The combination of learning vector quantization (LVQ) and NLP based hybrid models. The experiment is conducted using dataset collected from college students at Mansoura University. The proposed model provides accurate evaluations to improve student's writing skills.

3.6 Analytical Investigation 6

AI-6: How Do Different AES Systems and Evaluation Metrics Compare in Terms of Performance Metrics?

Evaluation metrics play a critical role in assessing the quality and reliability of Automatic Essay Scoring (AES) models. Among these, QWK is the most frequently utilized, offering a robust measure of agreement that disproportionately penalizes larger discrepancies between predicted and actual scores, thereby reflecting the severity of misclassifications more accurately. Pearson correlation coefficient is commonly employed to assess the strength and direction of the linear relationship between machine-predicted scores and human ratings, while Spearman's rank correlation coefficient evaluates the model's ability to preserve the ordinal ranking of essays, making it especially valuable when relative essay quality is prioritized over exact scores. Mean Squared Error (MSE) and Root Mean Squared Error (RMSE) provide measures of the average magnitude of prediction errors, with RMSE being

particularly interpretable due to its consistency with the original score scale, though both are less sensitive to ordinal relationships. In categorical scoring contexts, metrics such as Cohen's kappa and simple accuracy are also reported; however, they often fail to capture the subtleties of partial scoring errors. Additionally, measures like Mean Absolute Error (MAE) are sometimes utilized for their straightforward interpretability, and Earth Mover's Distance (EMD) has been proposed in some studies to better account for the ordered nature of essay scores. Recent research has also explored the use of Area under the Curve (AUC) for classification-based AES models and macro/micro F1-scores when multiple essay traits are scored simultaneously. The selection of an appropriate evaluation metric is crucial, as it directly influences the model development process and the interpretation of system performance in operational and research settings.

3.7 Analytical Investigation 7

AI-7: What Obstacles and Limitations Exist in the Current State of AES System Technology, and What Measures Can Be Taken to Overcome Them?

Many researchers have contributed their ideas in developing a robust AES system but still there exist many challenges.

Challenges Encountered and Solutions

1. **Context and Coherence**

Challenge: AES systems struggled to understand the logical flow and structure of essays, which is crucial for accurate grading.

Solution: This was addressed by refining the models to better understand how ideas are connected and ensuring the system could evaluate the overall coherence of the essay, not just individual sentences or words.

2. **Capturing Meaning Beyond Surface Features**

Challenge: Earlier systems focused on basic features like grammar and syntax, which didn't fully capture the intended meaning or creativity of the student's writing.

Solution: More sophisticated methods were introduced to focus on deeper aspects of meaning, such as the relationship between words and the context in which they were used.

3. **Limited Generalization Across Topics**

Challenge: Many AES systems were trained on narrow datasets, leading to poor generalization when faced with essays on unfamiliar topics.

Solution: The training process was broadened to include a wider range of topics and writing styles, making the system more adaptable to diverse educational settings.

4. **Data Imbalance**

Challenge: Certain essay prompts were overrepresented, leading to biased results that didn't reflect real-world writing diversity.

Solution: The datasets were augmented by generating additional essay samples or expanding the range of prompts, ensuring a more balanced and representative set of training data.

5. **Subjectivity and Inconsistency in Grading**

Challenge: Human graders often differ in their interpretations of the same essay, leading to inconsistencies in scoring.

Solution: Clearer grading rubrics and more consistent scoring criteria were developed to ensure that both human and automated systems applied similar standards for scoring essays.

The limitation that exists while going through the existing research work are enlightened below

1. ML based AES system, required handcrafted features. The most efficient feature must be tracked for efficient AES problem. The selection of features highly correlates the performance of AES problems.
2. Generally deep learning based AES system requires more training data. In AES tasks if the training data is limited for specific prompt in this case deep learning based algorithms fail to achieve good performance.
3. Fails in capturing the semantic meaning, intersection and dependencies among the words in sentences.
4. Lack in capturing complex grammatical structure of an essays.
5. Sentimental words are important while capturing debate type of questions. Some AES system fails in capturing such sentimental words in essays.
6. Domain independent type of essays is difficult to evaluate and gains the less performance in existing AES system.
7. Most AES system fails in providing the feedback back to students. Feedback plays an important role in evaluation process, by using feedback students can improve their individual performance.
8. Some AES systems are difficult to handle subjective topics or grading criteria leading to bias in evaluation or training data.
9. Table, diagram based component AES system are not in current use. The new researchers should focus on the deploying the integration these AES system in near future.

Despite the aforementioned limitations, AES systems possess the potential to serve as advantageous tools in consistent and accurate feedback, particularly in instances where the assessment of a large number of essays needs to be evaluated. Nevertheless, the careful utilization of these systems is crucial, as human evaluation remains essential for a comprehensive grasp of intricate writing tasks (Table 5).

Table 5 Different performance metrics used in AES tasks

Performance metrics—definition	Formula	Interpretation
Cohen's Kappa [64]: It is a Statistic that measures the level of agreement between two raters, taking into account the possibility of agreement occurring by chance	$K = \frac{PO-PE}{1-PE}$ *P*: Observed agreement (the proportion of cases where raters agree) *P*: Expected agreement by chance	The Kappa statistic ranges from − 1 to 1 Kappa = 1: Perfect agreement beyond chance. Kappa = 0: Agreement equivalent to chance. Kappa < 0: Agreement worse than chance
Quadratic Weighted Kappa is an extension of Cohen's Kappa that incorporates the degree of disagreement between categories	$K = 1 - \frac{\Sigma_i W_{i,j} O_{i,j}}{\Sigma_{i,i} W_{i,j}, E_{i,j}}$ k = Kappa values *Wi.j* = Weight matrices *Oi,j* = Observed matrices *Ei,j* = Expected matrices	The Kappa statistic ranges from –1 to 1 Kappa = 1: Perfect agreement beyond chance. Kappa = 0: Agreement equivalent to chance. Kappa < 0: Agreement worse than chance
Pearson correlation can be applied to evaluate the correlation between the scores assigned by the AES system and the scores provided by human raters	$r = x = \frac{\sum(x-mx)(y-my)}{\sqrt{\sum(x-mx)2(y-my)2}}$ *Xi* AES-assigned scores *yi*: Human-assigned scores : Mean of AES-assigned scores y^{x-}: Mean of human-assigned scores	The Pearson correlation coefficient ranges from –1 to 1: $r = 1$: Perfect positive correlation (as AES scores increase, human scores also increase) $r = -1$: Perfect negative correlation (as AES scores increase, human scores decrease) $r = 0$: No linear correlation
Root Mean Square Error (RMSE) can be applied when the system is providing continuous scores for essays, and you want to measure the accuracy of these scores compared to human-assigned scores	$RMSE = \sqrt{\sum_{i=1}^{n} \frac{\left(\hat{y}_i - y_i\right)^2}{n}}$ ^{y}i represents the mean (average) of the scores assigned by all raters *n* is the number of raters	Overall RMSE fails range of 0 to 100 A RMSE of 0 indicates that none of the predictions were correct A RMSE of 100 indicates that all predictions were correct
Mean Reciprocal Rank (MRR) involve assessing the quality of essays rather than ranking them	$\text{MRR} = \frac{1}{\lvert Q\rvert} \sum_{i=1}^{\lvert Q\rvert} \frac{1}{\text{rank}_i}$ *N* is the number of essays ^{rank}i is the rank of the first relevant essay for essay *i*	MRR score range from 0 to 1, with 1 being the best possible score
Fleiss Kappa can also be used when the same rater makes the measurement at more than two different times	$K = \frac{PO-PE}{1-PE}$ *P*: Observed agreement (the proportion of cases where raters agree) *P*: Expected agreement by chance	Fleiss' Kappa ranges from –1 to 1 A value of 1 indicates perfect agreement beyond chance 0 indicates agreement equivalent to chance. Negative values suggest agreement worse than chance

(continued)

Table 5 (continued)

Performance metrics—definition	Formula	Interpretation
Overall Accuracy in the context of AES refers to the proportionof essays that are correctly scored by the system out of the total number of essays in the dataset	Accuracy = $\frac{\text{Number of correct predciaction}}{\text{Total number of observations}}$	Overall Accuracy falls range of 0% to 100% An accuracy of 0% indicates that none of the predictions were correct An accuracy of 100% indicates that all predictions were correct
Spearman correlation coefficients can be used to quantify the relationship between the automated scores generated by the system and the human-assigned scores	$\rho = 1 - \frac{6\Sigma d_i^2}{n(n^2-1)}$ ρ = Spearman Correlation coefficient di = the difference in the ranks given to thetwo variables values for each item of the data **n** n = total number of observations	Pearson correlation, Spearman correlation ranges from –1 to 1: ρ = 1: Perfect positive monotonic correlation ρ = − 1: Perfect negative monotonic correlation ρ = 0: No monotonic correlation

3.8 Analytical Investigation 8

AI-8: What ethical considerations, such as privacy and transparency, need to be addressed in AES systems?

AES systems raise several ethical considerations that should be carefully addressed to mitigate the potential negative consequences of automated grading on student learning outcomes. Here are key ethical considerations associated with AES systems:

1. **Transparency and Accessibility**
   - Accessibility: AES systems should provide clear explanations of how they generate scores. This transparency helps educators and students understand the basis for the scores and promotes trust in the system.
   - Algorithmic Transparency: The algorithms used in AES should be transparent, and there should be disclosure about the features and criteria considered in the scoring process. This transparency helps in addressing concerns related to bias and fairness.
2. **Bias and Fairness**
   - Fairness in Assessment: Efforts should be made to identify and mitigate biases in AES systems to ensure fair assessment across diverse groups of students. This includes addressing potential biases related to ethnicity, gender, or socioeconomic status.
   - Regular Audits: Periodic audits of the AES system should be conducted to identify and rectify biases that may emerge over time. The system should be continuously monitored to ensure fairness.

3. **Privacy Concerns**
    - Data Privacy: AES systems often require access to a large amount of student data. It is crucial to handle this data with the utmost care, ensuring that personally identifiable information is protected and that data privacy regulations are adhered to.
    - Informed Consent: Students and educators should be informed about the data collection and usage policies associated with AES systems. Obtaining explicit consent for the use of student data is essential.
4. **Security**
    - **Data Security**: AES systems should implement robust security measures to protect the integrity and confidentiality of student data. This includes encryption, secure storage, and measures to prevent unauthorized access.
5. **Human Oversight**
    - **Human Review**: While AES systems automate the scoring process, human oversight is essential. There should be mechanisms for human review and intervention, particularly in cases where the automated scoring might be challenged or contested.
6. **Accessibility**
    - **Consideration for Diverse Learners**: AES systems should be designed with consideration for diverse learning styles and abilities. Ensure that the system does not disadvantage students with disabilities or those who may express themselves differently.
7. **Feedback and Improvement**
    - **Feedback Mechanisms**: Providing constructive feedback to students is a critical aspect of assessment. AES systems should be designed to offer meaningful feedback that aids learning and improvement.
    - **Continuous Improvement**: AES providers should commit to ongoing research and development to enhance the accuracy, fairness, and usefulness of their systems. Regular updates and improvements should be made based on feedback and emerging best practices.

3.9 Analytical Investigation 9

AI-9: What Are the Perceptions and Experiences of Educators and Students When Using AES for Grading and Feedback?

The perceptions and experiences of educators and students when using AES for grading and feedback can vary widely based on various factors, including the specific

technology used, the context of use, and the individuals involved. Here are some general points that capture a range of perspectives:

Educators

1. **Efficiency and Time-Saving**: Many educators appreciate the efficiency of AES in handling a large volume of essays quickly. It can save time compared to manual grading, allowing educators to focus on other aspects of teaching.
2. **Consistency**: AES can provide consistency in grading, as it applies predefined criteria consistently to all essays. This can help reduce subjective biases that may arise in manual grading.
3. **Quick Feedback**: Immediate feedback is a notable advantage of AES. Students can receive feedback promptly, enabling them to understand their strengths and weaknesses and make improvements.
4. **Supplemental Tool**: Some educators view AES as a valuable supplementary tool but may not rely solely on it. They might use it as an initial screening or for formative assessments, complemented by more in-depth manual grading for critical assignments.
5. **Concerns About Subjectivity**: Educators may express concerns about the potential subjectivity of AES and its ability to capture the nuances of complex writing. They may worry that it might not fully appreciate creativity, originality, or unconventional but valid approaches to a topic.

Students

1. **Immediate Feedback**: Students often appreciate the instant feedback provided by AES. It allows them to understand their performance quickly and work on areas that need improvement.
2. **Consistency**: Similar to educators, students may appreciate the consistency in grading. Knowing that essays are assessed using predefined criteria can provide a sense of fairness.
3. **Limited Personalization**: Some students may feel that AES lacks the personalization that human grading can offer. They might miss the detailed, individualized feedback and engagement that a teacher can provide.
4. **Concerns About Fairness**: There might be concerns among students about the fairness of AES, especially if the technology is perceived to be biased or if it does not adequately account for diverse writing styles.
5. **Preparation for Real-World Writing**: While AES may be efficient, some students and educators argue that manual grading better prepares students for real-world scenarios where human judgment is crucial.

3.10 Analytical Investigation 10

AI-10: What Are the Reported Research Outcomes in Automated Assessment Systems, and What Are the Next Steps for Educational Application?

Automated assessment systems, including AES and other forms of automated evaluation, have been extensively studied in research settings. Here are some reported research outcomes and trends, along with potential next steps for educational applications.

1. **Accuracy and Reliability**
   - **Research Outcome**: Studies have explored the accuracy and reliability of automated assessment systems in various domains. Many have found that well-designed systems can provide assessments that correlate well with human ratings.
   - **Next Steps**: Continuous improvement in the accuracy and reliability of these systems is crucial. Further research can focus on refining algorithms, addressing biases, and enhancing the ability to capture nuanced aspects of student performance.
2. **Personalized Feedback and Formative Assessment**
   - **Research Outcome**: One of the most valuable contributions of AES is its ability to provide immediate and personalized feedback to students. Traditional essay grading often takes time, leading to delayed feedback, which can reduce its effectiveness in helping students improve. With AES, students receive instant assessments, enabling them to identify their strengths and weaknesses in their writing immediately. AES systems can be fine-tuned to address specific writing features such as grammar, coherence, vocabulary use, and argument structure. This customization allows for feedback that is tailored to the individual student's needs. For example, if a student consistently struggles with coherence, the AES system can highlight these areas and suggest improvements, while students excelling in grammar may receive feedback focused on advancing their writing style or depth of argumentation. This feedback can help students understand their strengths and areas for improvement.
   - **Next Steps**: Enhancing the quality and specificity of feedback remains a priority. Research can focus on developing systems that offer actionable feedback tailored to individual student needs and learning objectives.
3. **Scalability and Efficiency**
   - **Research Outcome**: Automated assessment systems are valued for their scalability and efficiency, allowing quick evaluation of large volumes of student work.

- **Next Steps**: Future efforts can explore ways to further streamline assessment processes, reduce response times, and handle diverse types of assignments with efficiency.

4. **Ethical Considerations and Bias**
   - **Research Outcome**: Studies have highlighted the importance of addressing biases in automated assessment systems to ensure fairness across diverse student populations.
   - **Next Steps**: Continued research and development are needed to identify and mitigate biases. Attention should be given to the ethical implications of automated systems in educational contexts, including fairness, transparency, and accountability.
5. **Integration with Learning Management Systems (LMS)**
   - **Research Outcome**: Integration of automated assessment systems with Learning Management Systems has been explored to provide a seamless experience for educators and students.
   - **Next Steps**: Further research can focus on optimizing the integration process, ensuring interoperability with various LMS platforms, and addressing potential challenges in implementation.
6. **Human-AI Collaboration**
   - **Research Outcome**: Studies have emphasized the importance of human-AI collaboration in assessment, where automated systems support human decision-making rather than replace it entirely.
   - **Next Steps**: Future research can explore models of collaboration that leverage the strengths of both automated systems and human expertise, ensuring a balanced and effective approach to assessment.
7. **Adaptive Learning**
   - **Research Outcome**: AES can play a significant role in adaptive learning systems, which are designed to adjust content delivery based on a student's performance. With AES, learning platforms can analyze the scores and feedback generated for essays and adapt the next learning materials to better suit each student's needs. For instance, if an AES system identifies that a student struggles with writing introductions or thesis statements, the system can direct them to exercises or resources focused on these specific areas. If a student excels in one aspect, such as structure, the system may present more complex tasks or challenges in different writing aspects, such as argumentation or synthesis of ideas. This adaptability ensures that students are always working at the right level, neither overwhelmed by too difficult content nor under-challenged by overly simplistic tasks.
   - **Next Steps**: Continued efforts can focus on refining adaptive learning models, incorporating real- time assessment data, and personalizing educational content to meet the diverse needs of learners.

8. **Incorporating Multi-modal Assessment**
    - **Research Outcome**: Beyond text based assessment, there is increasing interest in incorporating multi-modal elements (e.g., images, audio) into automated assessment systems.
    - **Next Steps**: Future research can explore the development of robust multi-modal assessment systems that accurately evaluate a broader range of student expressions and skills.
9. **Improving Grading Accuracy**: Research should focus on enhancing the accuracy of AES systems, especially in capturing complex aspects of writing like argumentation, creativity, and critical thinking, which are often difficult for automated systems to assess.
10. **Handling Diverse Writing Styles**: AES systems need to be adapted to evaluate a wide variety of writing styles including informal language, non-native English, and creative writing. Addressing this will help systems better serve diverse student populations.

Research has yielded positive outcomes in the development and application of automated assessment systems, ongoing efforts are needed to address challenges, enhance system capabilities, and ensure ethical and equitable use in educational settings Collaborative interdisciplinary research involving educators, technologists, and policy makers will be crucial in shaping the future of automated assessment in education.

4 Conclusion

In conclusion, this review paper has presented a thorough examination of the current landscape of AES systems, offering insights into their key developments, methodologies, datasets, and the challenges they face. Through an analysis of ten critical investigative questions, the paper has highlighted important findings and trends that underscore the significant potential of AES systems in the education sector. These systems have advanced beyond traditional keyword-based approaches, integrating more complex algorithms that assess various elements such as syntax, semantic coherence, and sentiment-related terms, allowing for a more nuanced evaluation of student writing. Looking ahead, the future of AES research and application lies in addressing these challenges. There is a clear need for further works: First, the development of fair and unbiased models is crucial, and research should focus on identifying and mitigating any biases in automated scoring systems, particularly related to demographic factors like gender, ethnicity, or socioeconomic background. Second, transparency and interpretability in AES systems must be prioritized; future work could aim to create models that provide clear explanations of how scores are assigned, enhancing trust and understanding among users. Additionally, research should explore how AES can be designed to align with diverse learning

styles, including adjusting for different proficiency levels, writing approaches, and educational contexts. Another important direction is the exploration of ethical frameworks for the implementation of AES, ensuring these systems are deployed responsibly, with attention to privacy, fairness, and the potential psychological impacts on students. Furthermore, enhancing accessibility in AES systems, especially for institutions with limited resources, should be a priority, with efforts to create open-source tools or cost-effective solutions. Finally, longitudinal studies are needed to examine the long-term effects of AES on student learning and performance, particularly in terms of improving writing skills and providing actionable feedback for educators. These research directions are essential for ensuring that AES contributes positively to educational practices, will pave the way for a future where AES contributes meaningfully to educational assessment, providing valuable insights to both educators and students.

Summary of Survey

Aspect	Sample flow of details
Objective	To analyze methodologies, datasets, performance metrics, and challenges in developing robust AES systems for scalable, unbiased, and consistent essay evaluation
Key techniques	Statistical models, rule-based NLP methods, ML algorithms, deep learning (e.g., CNN, Bi-LSTM, transformers like BERT, RoBERTa), and hybrid approaches combining neural networks with handcrafted features
Datasets	Includes widely used datasets such as ASAP (12,978 essays), Mohler (2,273 answers), CLCFCE (323,192 sentences), AAE (402 persuasive essays), RTA (1,569 essays), Texas (2,442 responses), UKARA (Indonesian student essays), and others
Evaluation metrics	– Quadratic Weighted Kappa (QWK): Measures agreement with human raters – Root Mean Square Error (RMSE): Evaluates prediction accuracy – Pearson Correlation: Assesses alignment between human and AES scores
Challenges	– Scalability: Difficulties in grading large numbers of essays – Biases and Subjectivity: Rater fatigue, leniency, stereotypes, and halo effects – Domain Adaptation: Adapting models to diverse contexts and languages
Emerging trends	– Use of transformer models like BERT, RoBERTa, and GPT for better semantic understanding – Data augmentation techniques such as for generating synthetic essays – Explainable AI for transparency in scoring decisions
Applications	– Reducing human grading biases and inconsistencies – Providing instant feedback for better learning – Supporting low-resource languages and cross-domain assessments
Research contributions	– Incorporation of semantic similarity measures – Addressing coherence using discourse analysis and syntactic features – Hybrid models blending traditional NLP features with deep learning for enhanced reliability

References

1. Shermis, M.D., Burstein, J.C: Automated essay scoring: a cross-disciplinary perspective. Routledge (2003)
2. Jong, Y.J., Kim, Y.J., Ri, O.C.: Improving performance of automated essay scoring by using back-translation essays and adjusted scores. Math. Probl. Eng. **2022**(1), 6906587 (2022)
3. Ramesh, D., Sanampudi, S.K.: An automated essay scoring systems: a systematic literature review. Artif. Intell. Rev. **55**(3), 2495–2527 (2022)
4. Ling, G., Mollaun, P., Xi, X.: A study on the impact of fatigue on human raters when scoring speaking responses. Lang. Test. **31**(4), 479–499 (2014)
5. Leckie, G., Baird, J.A.: Rater effects on essay scoring: a multilevel analysis of severity drift, central tendency, and rater experience. J. Educ. Meas. **48**(4), 399–418 (2011)
6. Marchegiani, L., Reggiani, T., Rizzolli, M.: Loss averse agents and lenient supervisors in performance appraisal. J. Econ. Behav. Organ. **131**, 183–197 (2016)
7. Rangvid, B.S.: Gender discrimination in exam grading? Double evidence from a grading reform and a field experiment. VIVE working paper x (2018)
8. Kim, H.: Effects of rating criteria order on the halo effect in L2 writing assessment: a many-facet Rasch measurement analysis. Lang. Test. Asia **10**(1), 16 (2020)
9. Azizah, N., Suseno, M., Hayat, B.: Severity-leniency in writing assessment and its causal factors. KnE Soc. Sci. 173–185 (2020)
10. Brown, G.T., Abdulnabi, H.H.: Evaluating the quality of higher education instructor-constructed multiple-choice tests: Impact on student grades. In: Frontiers in Education, vol. 2, p. 24. Frontiers Media SA (2017)
11. Liu, J., Xu, Y., Zhu, Y.: Automated Essay Scoring Based on Two-Stage Learning. arXiv:1901.07744. (2019)
12. Dong, F., Zhang, Y., Yang, J.: Attention-based recurrent convolutional neural network for automatic essay scoring. In: Proceedings of the 21st Conference on Computational Natural Language Learning (CoNLL 2017), pp. 153–162 (2017)
13. Dasgupta, T., Naskar, A., Dey, L., Saha, R.: Augmenting textual qualitative features in deep convolution recurrent neural network for automatic essay scoring. In: Proceedings of the 5th Workshop on Natural Language Processing Techniques for Educational Applications, pp. 93–102. (2018)
14. Fowler Jr, F.J.: Survey research methods. Sage Publications (2013)
15. Bai, H., Hui, S.C.: A crowdsourcing-based incremental learning framework for automated essays scoring. Expert Syst. Appl. **238**, 121755 (2024)
16. Park, Y.H., Choi, Y.S., Park, C.Y., Lee, K.J.: EssayGAN: Essay data augmentation based on generative adversarial networks for automated essay scoring. Appl. Sci. **12**(12), 5803 (2022)
17. Li, J., Wu, J.: Automated essay scoring incorporating multi-level semantic features. In: International Conference on Artificial Intelligence in Education, pp. 206–211. Cham: Springer Nature Switzerland (2023)
18. Yamaura, M., Fukuda, I., Uto, M.: Neural automated essay scoring considering logical structure. In: International Conference on Artificial Intelligence in Education, pp. 267–278. Cham: Springer Nature Switzerland (2023)
19. Li, X., Yang, H., Hu, S., Geng, J., Lin, K., Li, Y.: Enhanced hybrid neural network for automated essay scoring. Expert. Syst. **39**(10), e13068 (2022)
20. Mizumoto, A., Eguchi, M.: Exploring the potential of using an AI language model for automated essay scoring. Res. Methods Appl. Linguist. **2**(2), 100050 (2023)
21. Sharma, A., Kabra, A., Kapoor, R.: Feature enhanced capsule networks for robust automatic essay scoring. In: Joint European Conference on Machine Learning and Knowledge Discovery in Databases, pp. 365–380. Cham: Springer International Publishing (2021)
22. Beseiso, M., Alzubi, O.A., Rashaideh, H.: A novel automated essay scoring approach for reliable higher educational assessments. J. Comput. High. Educ. **33**, 727–746 (2021)

23. Wu, Y., Henriksson, A., Nouri, J., Duneld, M., Li, X.: Retrieving key topical sentences with topic-aware BERT when conducting automated essay scoring. In: International Conference in Methodologies and intelligent Systems for Techhnology Enhanced Learning, pp. 123–132. Cham: Springer International Publishing (2022)
24. Ma, J., Li, X., Chen, M., Yang, W.: Enhanced hierarchical structure features for automated essay scoring. In: Information Retrieval: 27th China Conference, CCIR 2021, Dalian, China, October 29–31, 2021, Proceedings 27, pp. 168–179. Springer International Publishing (2021)
25. Sun, J., Song, T., Song, J., Peng, W.: Improving automated essay scoring by prompt prediction and matching. Entropy **24**(9), 1206 (2022)
26. Song, W., Zhang, K., Fu, R., Liu, L., Liu, T., Cheng, M.: Multi-stage pre-training for automated Chinese essay scoring. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP), pp. 6723–6733 (2020)
27. Wang, Y., Hu, R.: A prompt-independent and interpretable automated essay scoring method for Chinese second language writing. In: China National Conference on Chinese Computational Linguistics. pp. 450–470. Cham: Springer International Publishing (2021)
28. Jayadianti, H., Santosa, B., Cahyaning, J., Saifullah, S., Drezewski, R.: Essay auto-scoring using N-Gram and Jaro Winkler based Indonesian Typos. MATRIK: Jurnal Manajemen, Teknik Informatika dan Rekayasa Komputer **22**(2), 325–338 (2023)
29. Rajagede, R.A.: Improving automatic essay scoring for Indonesian language using simpler model and richer feature. Kinet.: Game Technol. Inf. Syst. Comput. Netw. Comput. Electron. Control. 11–18 (2021)
30. Wiratmo, A., Fatichah, C.: Indonesian short essay scoring using transfer learning dependency tree LSTM. Int. J. Intell. Eng. Syst. **13**(2) (2020)
31. Singh, S., Pupneja, A., Mital, S., Shah, C., Bawkar, M., Gupta, L.P., Kumar, A., Kumar, Y., Gupta, R., Shah, R.R.: H-AES: towards automated essay scoring for Hindi. Proc. AAAI Conf. Artif. Intell. **37**(13), 15955–15963 (2023)
32. Meccawy, M., Bayazed, A.A., Al-Abdullah, B., Algamdi, H.: Automatic essay scoring for Arabic short answer questions using text mining techniques. Int. J. Adv. Comput. Sci. Appl. **14**(6) (2023)
33. Bramhe, S., Pathak, S.S.: Robotic surgery: a narrative review. Cureus **14**(9) (2022)
34. Page, E.B.: Project Essay Grade: PEG (2003)
35. Golubovich, J., Chatterjee, D., Robbins, S.: Educational Testing Service (ETS) Job Profiler for Validity Studies: Overview and Planned Application (2015)
36. Foltz, P.W., Laham, D., Landauer, T.K.: The intelligent essay assessor: applications to educational technology. Interact. Multimed. Electron. J. Comput.-Enhanc. Learn. **1**(2), 939–944 (1999)
37. Rudner, L.M., Garcia, V., Welch, C.: An evaluation of IntelliMetric™ essay scoring system. J. Technol. Learn. Assess. **4**(4) (2006)
38. Attali, Y., Burstein, J.: Automated essay scoring with e-rater® V. 2. J. Technol. Learn. Assess. **4**(3) (2006)
39. Rudner, L. M., Liang, T.: Automated essay scoring using Bayes' theorem. Technol. Learn. Assess. **1**(2) (2002)
40. Laham, D., Foltz, P. W., Landauer, T.: The intelligent essay assessor. IEEE Intell. Syst. (2000)
41. Zupanc, K., Bosnic, Z.: Advances in the field of automated essay evaluation. Informatica **39**(4) (2016)
42. Burstein, J., Tetreault, J., Madnani, N.: The e-rater® automated essay scoring system. In: Handbook of Automated Essay Evaluation, pp. 55–67. Routledge (2013)
43. Leacock, C., Chodorow, M.: C-rater: automated scoring of short-answer questions. Comput. Humanit. **37**, 389–405 (2003)
44. Dikli, S.: An overview of automated scoring of essays. J. Technol. Learn. Assess. **5**(1) (2006)
45. Learning, V.: MY Access! efficacy report. Newtown, PA: Vantage Learning. Retrieved 6 Aug 2009 (2007)
46. Rus, V., Lintean, M., Banjade, R., Niraula, N.B., Stefanescu, D.: Semilar: The semantic similarity toolkit. In: Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics: System Demonstrations, pp. 163–168 (2013)

47. Batanović, V., Cvetanović, M., Nikolić, B.: Fine-grained semantic textual similarity for serbian. In: Proceedings of the Eleventh International Conference on Language Resources and Evaluation (LREC 2018) (2018)
48. Han, L., Kashyap, A.L., Finin, T., Mayfield, J., Weese, J.: UMBC_EBIQUITY-CORE: Semantic textual similarity systems. In: Second Joint Conference on Lexical and Computational Semantics (*SEM), Vol. 1: Proceedings of the Main Conference and the Shared Task: Semantic Textual Similarity, pp. 44–52 (2013)
49. Bhingardive, S., Redkar, H., Sappadla, P., Singh, D., Bhattacharyya, P.: IndoWordNet: similarity-computing semantic similarity and relatedness using IndoWordNet. In: Proceedings of the 8th Global WordNet Conference (GWC), pp. 39–43 (2016)
50. Mazandu, G.K., Chimusa, E.R., Mulder, N.J.: Gene ontology semantic similarity tools: survey on features and challenges for biological knowledge discovery. Brief. Bioinform. **18**(5), 886–901 (2017)
51. Kolb, P.: Disco: A multilingual database of distributionally similar words. In: Proceedings of KONVENS, vol. 156, p. 8 (2008)
52. Faria, D., Pesquita, C., Couto, F.M., Falcão, A.: Proteinon: A Web Tool for Protein Semantic Similarity (2007)
53. Ranawella, T.C.: Turning Usage and Its Impact on Academic Integrity and Honesty of General Sir John Kotelawala Defence University (KDU) (2018)
54. Maulidina, P., Wibowo, H.: The use of Grammarly tools to enrich student's writing ability. Lingua **18**(2), 179–189 (2022)
55. Weber-Wulff, D.: Plagiarism detection software: Promises, pitfalls, and practices. Handb. Acad. Integr. **625** (2016)
56. Milo, T., Somech, A., Youngmann, B.: Semsim: Combining Structural and Semantic Similarity (full version)
57. Speer, R., Lowry-Duda, J.: Luminoso at Semeval-2018 Task 10: Distinguishing Attributes Using Text Corpora and Relational Knowledge (2018). arXiv:1806.01733
58. Farag, Y., Yannakoudakis, H., Briscoe, T.: Neural Automated Essay Scoring and Coherence Modeling for Adversarially Crafted Input (2018). arXiv:1804.06898
59. Li, C., Lin, L., Mao, W., Xiong, L., Lin, Y.: An automated essay scoring model based on stacking method. In: 2022 IEEE 2nd International Conference on Software Engineering and Artificial Intelligence (SEAI), pp. 248–252. IEEE (2022)
60. Mohler, M., Bunescu, R., Mihalcea, R.: Learning to grade short answer questions using semantic similarity measures and dependency graph alignments. In: Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies, pp. 752–762 (2011)
61. Yannakoudakis, H., Briscoe, T., Medlock, B.: A new dataset and method for automatically grading ESOL texts. In: Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies, pp. 180–189 (2011)
62. Schmalz, V.J., Brutti, A.: Automatic assessment of English CEFR levels using BERT embeddings. In: Proceedings of the Eighth Italian Conference on Computational Linguistics (2021)
63. Dzikovska, M.O., Nielsen, R., Brew, C., Leacock, C., Giampiccolo, D., Bentivogli, L., Clark, P., Dagan, I, Dang, H.T.: Semeval-2013 task 7: The joint student response analysis and 8th recognizing textual entailment challenge. In: Second Joint Conference on Lexical and Computational Semantics (*SEM), Vol. 2: Proceedings of the Seventh International Workshop on Semantic Evaluation (SemEval 2013), pp. 263–274 (2013)
64. Galhardi, L.B., de Mattos Senefonte, H.C., de Souza, R.C.T., Brancher, J.D.: Exploring distinct features for automatic short answer grading. In: Encontro Nacional de Inteligência Artificial e Computacional (ENIAC), pp. 1–12. SBC (2018)
65. Gomaa, W.H., Fahmy, A.A.: Short answer grading using string similarity and corpus-based similarity. Int. J. Adv. Comput. Sci. Appl. (IJACSA) **3**(11) (2012)
66. Groth, R.E.: Applying design-based research findings to improve the Common Core State Standards for Data and Statistics in Grades 4–6. J. Stat. Educ. **27**(1), 29–36 (2019)

67. Qi, H., Wang, Y., Dai, J., Li, J., Di, X.: Attention-based hybrid model for automatic short answer scoring. In: Simulation Tools and Techniques: 11th International Conference, SIMUtools 2019, Chengdu, China, July 8–10, 2019, Proceedings 11, pp. 385–394. Springer International Publishing (2019)
68. Parra, G.L., Calero, S.X.: Automated writing evaluation tools in the improvement of the writing skill. Int. J. Instr. **12**(2), 209–226 (2019)
69. Kumar, S., Chakrabarti, S., Roy, S.: Earth mover's distance pooling over siamese LSTMs for automatic short answer grading (2017)
70. Birla, N., Jain, M.K., Panwar, A.: Automated assessment of subjective assignments: a hybrid approach. Expert Syst. Appl. **203**, 117315 (2022)
71. Ramnarain-Seetohul, V., Bassoo, V., Rosunally, Y.: Work-in-progress: computing sentence similarity for short texts using transformer models. In: 2022 IEEE Global Engineering Education Conference (EDUCON), pp. 1765–1768. IEEE (2022)
72. Beseiso, M., Alzahrani, S.: An empirical analysis of BERT embedding for automated essay scoring. Int. J. Adv. Comput. Sci. Appl. **11**(10) (2020)
73. Xue, J., Tang, X., Zheng, L.: A hierarchical BERT-based transfer learning approach for multidimensional essay scoring. IEEE Access **9**, 125403–125415 (2021)
74. ElNaka, A., Nael, O., Afifi, H., Sharaf, N.: AraScore: investigating response-based Arabic short answer scoring. Procedia Comput. Sci. **189**, 282–291 (2021)
75. Wang, Y., Wang, C., Li, R., Lin, H.: On the use of Bert for automated essay scoring: Joint learning of multi-scale essay representation (2022) arXiv:2205.03835
76. Janda, H.K., Pawar, A., Du, S., Mago, V.: Syntactic, semantic and sentiment analysis: the joint effect on automated essay evaluation. IEEE Access **7**, 108486–108503 (2019)
77. Xia, L., Liu, J., Zhang, Z.: Automatic essay scoring model based on two-layer bi-directional long-short term memory network. In: Proceedings of the 2019 3rd International Conference on Computer Science and Artificial Intelligence, pp. 133–137 (2019)
78. Rodriguez, P.U., Jafari, A., Ormerod, C.M.: Language Models and Automated Essay Scoring (2019) arXiv:1909.09482
79. Liang, G., On, B.W., Jeong, D., Kim, H.C., Choi, G.S.: Automated essay scoring: a siamese bidirectional LSTM neural network architecture. Symmetry **10**(12), 682 (2018)
80. Cozma, M., Butnaru, A.M., Ionescu, R.T.: Automated essay scoring with string kernels and word embeddings (2018) arXiv:1804.07954
81. Song, S., Zhao, J.: Automated Essay Scoring Using Machine Learning. Stanford University (2013)
82. Ramalingam, V.V., Pandian, A., Chetry, P., Nigam, H.: Automated essay grading using machine learning algorithm. In: Journal of Physics: Conference Series, vol. 1000, p. 012030. IOP Publishing (2018)
83. Elalfi, A.E.E., Elgamal, A.F., Amasha, N.A.: Automated essay scoring using Word2vec and support vector machine. Int. J. Comput. Appl. **177**(25), 975–8887 (2019)
84. Wu, L.: Automatic English essay scoring algorithm based on machine learning. In 2023 IEEE International Conference on Integrated Circuits and Communication Systems (ICICACS), pp. 1–4. IEEE (2023)
85. Azahar, M.Z.A.B., Ghauth, K.I.B.: A Hybrid automated essay scoring using NLP and random forest regression. In: International Conference on Computer, Information Technology and Intelligent Computing (CITIC 2022), pp. 448–457. Atlantis Press (2022)
86. Dadi, R., Sanampudi, S.K.: Coherence Based Automatic Short Answer Scoring Using Sentence Embedding (2023)
87. Kumar, V., Boulanger, D.: Explainable automated essay scoring: Deep learning really has pedagogical value. In: Frontiers in education, vol. 5, pp. 572367. Frontiers Media SA (2020)
88. Cai, C.: Automatic essay scoring with recurrent neural network. In: Proceedings of the 3rd International Conference on High Performance Compilation, Computing and Communications, pp. 1–7 (2019)
89. Shibata, T., Uto, M.: Analytic automated essay scoring based on deep neural networks integrating multidimensional item response theory. In: Proceedings of the 29th International Conference on Computational Linguistics, pp. 2917–2926 (2022)

90. Tashu, T.M., Maurya, C.K., Horvath, T.: Deep Learning Architecture for Automatic Essay Scoring (2022) arXiv preprint arXiv:2206.08232
91. Lu, C., Cutumisu, M.: Integrating Deep Learning into an Automated Feedback Generation System for Automated Essay Scoring. International Educational Data Mining Society (2021)
92. Zhu, W., Sun, Y.: Automated essay scoring system using multi-model machine learning. In: CS & IT Conference Proceedings, vol. 10(12). CS & IT Conference Proceedings (2020)
93. Dhini, B.F., Girsang, A.S.: Development of an automated scoring model using sentence transformers for discussion forums in online learning environments. J. Comput. Inf. Technol. **30**(2), 85–99 (2022)
94. Wang, J., Chen, J., Ou, X., Han, Q., Tang, Z.: Multi-level Feature Fusion for Automated Essay Scoring (2023)
95. Zhang, P.: Long-former-based Automated Writing Assessment for English Language Learners (2020)
96. Xu, W., Mahmud, R. B., Lam, H.W.: A systematic literature review: are automated essay scoring systems competent in real-life education scenarios? IEEE Access (2024)
97. Daniels, P.: Automated grammar scoring with AI chatbots. TEFL Prax. J. **2**(1), 59–68 (2023)
98. Prabhu, S., Akhila, K., Sanriya, S.: A hybrid approach towards automated essay evaluation based on Bert and feature engineering. In 2022 IEEE 7th International Conference for Convergence in Technology (I2CT), pp. 1–4. IEEE (2022)
99. Uto, M., Xie, Y., Ueno, M.: Neural automated essay scoring incorporating handcrafted features. In: Proceedings of the 28th International Conference on Computational Linguistics, pp. 6077–6088 (2020)
100. Jin, C., He, B., Hui, K., Sun, L.: TDNN: a two-stage deep neural network for prompt-independent automated essay scoring. In: Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics, vol. 1: Long Papers, pp. 1088–1097 (2018)
101. Kuyoro, S.O., Eluwa, J.M., Akinsola, J.E.T., Ayankoya, F.Y., Omotunde, A.A., Adegbenjo, A.A.: Intelligent Essay Grading System using Hybrid Text Processing Techniques. Int. J. Sci. Res. Comput. Sci. Eng. Inf. Technol. (IJSRCSEIT) (2020)
102. Chen, Z., Quan, Y., Qian, D.: Automatic Essay Scoring Model Based on Multi-channel CNN and LSTM. In Intelligent Computing and Block Chain: First BenchCouncil International Federated Conferences, FICC 2020, Qingdao, China, October 30 –November 3, 2020, Revised Selected Papers 1, pp. 337–346. Springer Singapore (2021)
103. Shehab, A., Elhoseny, M., Hassanien, A.E.: A hybrid scheme for automated essay grading based on LVQ and NLP techniques. In 2016 12th International Computer Engineering Conference (ICENCO), pp. 65–70. IEEE (2016)